

Signature-Based Models in Finance



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Abstract We consider two classes of asset price models where either the price or the volatility dynamics are described by a linear function of the (time extended) signature of a primary process, in general a multidimensional continuous semimartingale. These model classes are universal in the sense that classical models can be approximated arbitrarily well or are simply nested in our setup. Under the additional assumption that the primary process is polynomial, we obtain tractable option pricing formulas for so-called sig-payoffs in the first class and closed form expressions for the VIX squared and the log-price in the second one. In both cases the signature samples can be easily precomputed, hence the calibration task can be split into an offline sampling and a standard optimization. We present several applications, in particular the successfully solved joint SPX/VIX calibration problem.

1 Introduction

The present chapter is based on the two articles [13] and [11].

In recent years, the traditional approach of calibrating a few well-interpretable parameters has given way to learning the model's characteristics as a whole, by leveraging all available sources of data. Consequently, overparametrized models have become increasingly important. This shift has opened the door to more robust

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and data-driven model selection mechanisms. At the same time the choice of admissible models must, however, be limited to those adhering to first principles such as “no-arbitrage”. Finding classes of dynamic processes that are data-driven *and* satisfy such well-established theoretical principles can be achieved by relying on different universal approximation theorems.

One class of such financial models are so-called *signature SDEs*, as considered in [12, 37]. These are Itô-diffusions, where the characteristics are linear or real-analytic functions of the *signature* of some (properly extended) driving Brownian motion or the process itself. The models that we consider in this chapter can be embedded in the framework of signature SDEs and serve as particularly tractable examples thereof.

In the first part (Sect. 3) of the present chapter our goal is to provide a data-driven, universal, tractable and easy to calibrate asset price model. For the sake of exposition we shall assume throughout that the asset S is one-dimensional. We model S as a linear function of the signature of—what we call—*primary process*. This primary process can either be a classical driving signal, e.g., a Brownian motion or a polynomial process [9], but also a general d -dimensional continuous semimartingale $X = (X_t^1, \dots, X_t^d)_{t \geq 0}$, usually augmented with time t . The corresponding time-extended process is denoted by $(\widehat{X}_t)_{t \geq 0} = (t, X_t^1, \dots, X_t^d)_{t \geq 0}$ and its signature by $\widehat{\mathbb{X}}$. The asset S is then modeled/approximated via a process $S_n(\ell)$ defined as

$$S_n(\ell)_t := \ell(\widehat{\mathbb{X}}_t), \quad (1)$$

where ℓ is a linear map of the signature of $\widehat{X}$ up to some level of truncation $n \in \mathbb{N}$ which needs to be inferred from data (see Definition 3.2 for further details). Note that the parameters of X are prespecified beforehand and can thus be seen—in analogy to machine learning terminology—as *hyperparameters*. This is crucial as it allows to split the calibration task into precomputable samples of the signature and a standard optimization to find the parameters of the linear map. This is one of the attractive features, further ones are summarized subsequently.

No Arbitrage We will present in Sect. 3.2, that (1) can also be expressed in terms of stochastic integrals, whence is straightforward to deduce no-arbitrage conditions.

Universality We refer the reader to [11, Example 3.15], where we show that classical stochastic volatility models with sufficiently regular coefficients can be arbitrarily well approximated by models of form (1).

Option Pricing Formulas for Sig-Payoffs By approximating (path dependent) payoffs via so-called sig-payoffs of the form $L(\widehat{\mathbb{S}}_T(\ell))$ (see also [34]), where $\widehat{\mathbb{S}}$ denotes the signature of $(\widehat{S}_t)_t := (t, S_t)_t$ and L a linear map, we show that this kind of approximate option pricing reduces to the computation of the expected signature of $\widehat{X}$. These formulas lead to high tractability whenever $\mathbb{E}[\widehat{\mathbb{X}}_T]$ is easy to compute. This is the case for all polynomial process as analyzed in [12] and in Sect. 2 for the truncated signature.

Calibration to Options As mentioned above, when calibrating to the market's volatility surface (see Sect. 3.5), we precompute Monte Carlo samples of $\widehat{\mathbb{X}}$ and are then only left with finding the parameters ℓ , which is subject to a standard optimization. We perform for both simulated and real market data (S&P 500 index) a full calibration to the volatility surface, and show that this is not only highly accurate but also very fast (see [11] for details, in particular in the case of time dependent parameters).

The second part (Sect. 4) treats a slightly different class of signature-based models and constitutes a novel contribution for joint VIX and S&P 500 modeling as well as calibration. In the context of volatility modeling the joint calibration to SPX¹ and VIX options is still considered a rather hard problem which has become increasingly important over the past years. However it is worth to mention that significant progress has been made recently (see [13] and [17]). We address the reader to [13, 17, 28] for an extensive literature review.

Inspired by the previous model, we now study a stochastic volatility model for the discounted price process $S = (S_t)_{t \geq 0}$, namely

$$dS_t(\ell) = S_t(\ell)\sigma_t^S(\ell)dB_t,$$

with some initial condition $S_0 \in \mathbb{R}_+$, standard Brownian motion B , and volatility process σ^S satisfying

$$\sigma_t^S(\ell) := \ell(\widehat{\mathbb{X}}_t), \quad (2)$$

for a linear map ℓ of the signature $\widehat{\mathbb{X}}_t$ of a time extended d -dimensional continuous semimartingale X , which takes here the role of the primary process. We thus assume that the signature of $\widehat{X}$ serves as a linear regression basis for the volatility process, while the parameters of the linear map ℓ have to be learned from (option price) data.

Also in this case the modeling framework can be seen as *universal* in the class of continuous non-rough stochastic volatility, which is a consequence of the universal approximation properties of the signature. Besides that it truly nests several classical models (see Remark 4.3). and incorporates both, purely Markovian (in (S, X)) and path-dependent ones. Moreover, it provides for the first time a signature-based approach for pricing VIX options and highly accurate joint calibration results, as illustrated in Sect. 4.2. For the latter we exploit the following mathematical and numerical properties.

- Setting $Z := (X, B)$, then not only $\sigma^S(\ell)$ but also the log-price $\log(S(\ell))$ is a linear function of the signature of $\widehat{Z}$. Therefore no (Euler) simulation scheme

¹ Note that the SPX is a theoretical index tracking the value of the S&P 500 without holding an actual portfolio of the constituent stocks. In the following we shall use the terms SPX and S&P 500 interchangeably.

is needed to sample the price process, leading to immediate computational advantages.

- If we additionally assume X to be a polynomial process (see Definition 2.1 and [9, 19]), we obtain an analytic expression for the VIX, which only involves the computation of a matrix exponential. This follows from the property that the truncated signature of a polynomial process is again a polynomial process (see Sect. 2).
- We can apply a Monte Carlo approach for option pricing *and* calibration where we are able to generate the signature samples of $\hat{Z}$ offline and independently of the model parameters to be optimized. As in the previous model the calibration task can thus be split into an offline sampling procedure and a standard optimization, since the latter does not require re-sampling for updated model parameters ℓ .

The codes for the first part are available at [GuidoGazzani-ai/sigsde_calibration](#), while an implementation of the model for the joint calibration can be found in [GuidoGazzani-ai/_jointcalib_sigsde](#) or [janka-moeller/joint_calib_SPX_VIX](#). Moreover another implementation of the expected signature of polynomial process is available at [github.com/sarasvaluto/AffPolySig](#). The first repository refers to [11], the other three have been introduced in [13]. We additionally share a [Colab notebook](#) where the reader can benefit from an implementation of the building blocks of signature-based models as here introduced.

1.1 Notation

Let us first fix the notation used throughout the chapter. Recall the notions introduced in chapter “A Primer on the Signature Method in Machine Learning”, in particular the *algebra of formal power series* which we will also refer to as *extended tensor algebra* and denote by $T((\mathbb{R}^d))$. For a multi-index $I := (i_1, \dots, i_n) \in \{1, \dots, d\}^n$ we set $|I| := n$. We also consider the empty index $I := \emptyset$ and set $|I| := 0$. For each index I we then define

$$I' := \begin{cases} (i_1, \dots, i_{|I|-1}) & \text{if } |I| \geq 2, \\ \emptyset & \text{if } |I| = 1, \\ 0 & \text{if } |I| = 0. \end{cases} \quad (3)$$

Similarly we set $I'' := (I')'$ if $|I| \neq 0$ and $I'' = 0$ if $|I| = 0$. Let us denote by $e_\emptyset$ the basis element corresponding to $(\mathbb{R}^d)^{\otimes 0}$, then each element $\mathbf{a} \in T((\mathbb{R}^d))$ can be expressed as

$$\mathbf{a} = \sum_{|I| \geq 0} \mathbf{a}_I e_I,$$

for a collection of $\mathbf{a}_I \in \mathbb{R}$. Moreover a vectorization of the elements of the truncated tensor algebra $T^{(n)}(\mathbb{R}^d)$ whose elements are of the form

$$\mathbf{u} = \sum_{0 \leq |I| \leq n} \mathbf{u}_I e_I$$

will prove useful. To this end, we introduce the isomorphism $\mathbf{vec} : T^{(n)}(\mathbb{R}^d) \rightarrow \mathbb{R}^{d_n}$ and an arbitrary but fixed injective labeling function $\mathcal{L} : \{I : |I| \leq n\} \rightarrow \{1, \dots, d_n\}$, such that

$$\mathbf{vec}(\mathbf{u}) := \sum_{|I| \leq n} e_{\mathcal{L}(I)} \mathbf{u}_I, \quad (4)$$

with $d_n := \frac{d^{n+1}-1}{d-1}$.

We stick to the convention of mathematical finance, that the process $S = (S_t)_{t \geq 0}$ denotes the price process. Therefore, we denote the signature of a process X by $\mathbb{X}$. Moreover, for a multi-index I and $0 \leq s \leq t$, we denote the corresponding increment of a component of the signature by its projection on e_I that is $\langle e_I, \mathbb{X}_{s,t} \rangle$. To make the connection to the notation introduced in chapter “A Primer on the Signature Method in Machine Learning”: for a multi-index I and times $0 \leq s \leq t$ we denote the corresponding element of the signature of a process X by

$$\langle e_I, \mathbb{X}_{s,t} \rangle = S(X)_{s,t}^I,$$

and write $\mathbb{X}_t := \mathbb{X}_{0,t}$. We emphasize again that from now on we will only use the letter S to denote the price process and never the signature of a process. Recall that for any $t \geq 0$ the signature of an $\mathbb{R}^d$ -dimensional process $X = (X_t)_{t \geq 0}$ it holds that $\mathbb{X}_t \in T((\mathbb{R}^d))$ at any time $t \geq 0$. We denote its canonical projection to $T^{(n)}(\mathbb{R}^d)$ by $\mathbb{X}^n$ and refer to it as the *truncated signature*.

In this chapter, we assume the stochastic processes to be continuous semimartingales. Recall that the elements of the signature of a continuous semimartingale $X = (X_t)_{t \geq 0}$ can be defined recursively as

$$\langle e_\emptyset, \mathbb{X}_{s,t} \rangle := 1, \quad \langle e_I, \mathbb{X}_{s,t} \rangle := \int_s^t \langle e_{I'}, \mathbb{X}_{s,r} \rangle \circ dX_r^{i_n},$$

for each $I = (i_1, \dots, i_n)$, $I' = (i_1, \dots, i_{n-1})$ and $0 \leq s \leq t$, where $\circ$ denotes the Stratonovich integral.

Definition 1.1 (Shuffles) Set $I = (i_1, \dots, i_n)$ and $J = (j_1, \dots, j_m)$ for some $n \in \mathbb{N}$ and $m > 0$. We define the shuffle $\sqcup$ and half-shuffle $\tilde{\sqcup}$ as $e_I \sqcup e_\emptyset := e_I$, $e_I \tilde{\sqcup} e_\emptyset := 0$, respectively and

$$e_I \sqcup e_J := (e_{I'} \sqcup e_J) \otimes e_{i_n} + (e_I \sqcup e_{J'}) \otimes e_{j_m}, \quad e_I \tilde{\sqcup} e_J := (e_I \sqcup e_{J'}) \otimes e_{j_m}.$$

Remark 1.2 Using shuffle and half-shuffle one can recover linear representations of more complex operations on the signature. In particular, for any pair of multi-indices I, J and each $t \geq 0$ it holds that (see e.g. [11])

$$\langle e_I, \mathbb{X}_t \rangle \langle e_J, \mathbb{X}_t \rangle = \langle e_I \sqcup e_J, \mathbb{X}_t \rangle \quad \text{and} \quad \int_0^t \langle e_I, \mathbb{X}_s \rangle \circ d \langle e_J, \mathbb{X}_s \rangle = \langle e_I \widetilde{\sqcup} e_J, \mathbb{X}_t \rangle.$$

Choosing for example $I = (1)$ and $J = (2)$ we get that

$$e_{(1)} \sqcup e_{(2)} = (e_{\emptyset} \sqcup e_{(2)}) \otimes e_{(1)} + (e_{(1)} \sqcup e_{\emptyset}) \otimes e_{(2)} = e_{(2,1)} + e_{(1,2)}$$

and $e_{(1)} \widetilde{\sqcup} e_{(2)} = (e_{(1)} \sqcup e_{\emptyset}) \otimes e_{(2)} = e_{(1,2)}$. Setting $X_0 = 0$ this corresponds to

$$\langle e_{(1)}, \mathbb{X}_t \rangle \langle e_{(2)}, \mathbb{X}_t \rangle = X_t^1 X_t^2 = \int_0^t X_s^2 \circ dX_s^1 + \int_0^t X_s^1 \circ dX_s^2 = \langle e_{(2,1)} + e_{(1,2)}, \mathbb{X}_t \rangle$$

and $\int_0^t \langle e_{(1)}, \mathbb{X}_s \rangle \circ d \langle e_{(2)}, \mathbb{X}_s \rangle = \int_0^t X_s^1 \circ dX_s^2 = \langle e_{(1,2)}, \mathbb{X}_t \rangle$, respectively.

2 Expected Signature of Polynomial Processes

Consider the $\mathbb{R}^d$ -valued process $Y = (Y_t)_{t \geq 0}$ given by

$$dY_t = b(Y_t)dt + \sqrt{a(Y_t)}dW_t, \quad Y_0 = y_0, \quad (5)$$

for some d -dimensional Brownian motion W and some maps $a : \mathbb{R}^d \rightarrow \mathbb{S}_+^d$ (where $\mathbb{S}_+^d$ denotes the set of symmetric positive definite $d \times d$ matrices) and $b : \mathbb{R}^d \rightarrow \mathbb{R}^d$. Set $\sigma(Y_t) := \sqrt{a(Y_t)}$. Assume then that $Y = (Y_t)_{t \geq 0}$ is a polynomial process as defined in the next definition and denote by $\mathbb{Y}$ the corresponding signature.

Definition 2.1 (Polynomial Process) A process $Y = (Y_t)_{t \geq 0}$ satisfying (5) is called polynomial process if a_{ij} is a polynomial of degree at most 2 and b_j is a polynomial of degree at most 1 for each $i, j \in \{1, \dots, d\}$.

Various representations of the conditional expected signature of Y and analogous quantities, in particular for Brownian motion, can be found in [3, 5, 18, 32, 33]. We also refer to [15, 20] for the case of Lévy processes. Our approach aligns with [12] and is grounded in the classical theory of polynomial processes (see [9] and [19]). Although the framework of general signature SDEs considered in [12] requires results for infinite-dimensional stochastic processes (see, for example, [8, 14]), the current assumption of Y being a polynomial process allows us to remain within the finite-dimensional setting.

Lemma 2.2 *Let $(Y_t)_{t \geq 0}$ be a polynomial process satisfying (5). The corresponding drift and diffusion coefficients b and a can then be written as*

$$b_j(y) = b_j^c + \sum_{k=1}^d b_j^k y_k \quad \text{and} \quad a_{ij}(y) = a_{ij}^c + \sum_{k=1}^d a_{ij}^k y_k + \sum_{k,h=1}^d a_{ij}^{kh} y_k y_h,$$

for some $b_j^c, b_j^k, a_{ij}^c, a_{ij}^k, a_{ij}^{kh} = a_{ij}^{hk} \in \mathbb{R}$, with $i, j = 1, \dots, d$. Moreover,

$$b_j(Y_t) = \langle \mathbf{b}_j, \mathbb{Y}_t^1 \rangle \quad \text{and} \quad a_{ij}(Y_t) = \langle \mathbf{a}_{ij}, \mathbb{Y}_t^2 \rangle$$

for

$$\begin{aligned} \mathbf{b}_j &= \left(b_j^c + \sum_{k=1}^d b_j^k Y_0^k \right) e_\emptyset + \sum_{k=1}^d b_j^k e_k \quad \text{and} \\ \mathbf{a}_{ij} &= \left(a_{ij}^c + \sum_{k=1}^d a_{ij}^k Y_0^k + \sum_{k,h=1}^d a_{ij}^{kh} Y_0^k Y_0^h \right) e_\emptyset + \sum_{k=1}^d \left(a_{ij}^k + 2 \sum_{h=1}^d a_{ij}^{kh} Y_0^h \right) e_k \\ &\quad + \sum_{k,h=1}^d a_{ij}^{kh} e_k \sqcup e_h. \end{aligned}$$

Note that the upper index on Y_0^k and Y_0^h refers to the components of Y and not to powers.

Proof The first representation follows by the definition of polynomial processes, according to which b and a are polynomials of degree at most 1 and 2, respectively. For the second representation it then suffices to note that $\langle e_\emptyset, \mathbb{Y}_t^1 \rangle = \langle e_\emptyset, \mathbb{Y}_t^2 \rangle = 1$, $\langle e_k, \mathbb{Y}_t^1 \rangle = \langle e_k, \mathbb{Y}_t^2 \rangle = (Y_t^k - Y_0^k)$, and $\langle e_k \sqcup e_h, \mathbb{Y}_t^2 \rangle = (Y_t^k - Y_0^k)(Y_t^h - Y_0^h)$. $\square$

Lemma 2.3 Let $(Y_t)_{t \geq 0}$ be a polynomial process satisfying (5). Then the truncated signature $(\mathbb{Y}_t^n)_{t \geq 0}$ is a polynomial process and for each $|I| \leq n$ it holds that

$$\langle e_I, \mathbb{Y}_t^n \rangle = \int_0^t \langle L e_I, \mathbb{Y}_s^n \rangle ds + \int_0^t \langle e_{I'}, \mathbb{Y}_s^n \rangle \sigma_{i_{|I|}}(Y_s) dW_s,$$

where $L : T((\mathbb{R}^d)) \rightarrow T((\mathbb{R}^d))$ satisfies $L(T^{(n)}(\mathbb{R}^d)) \subseteq T^{(n)}(\mathbb{R}^d)$ and is given by

$$L e_I = e_{I'} \sqcup \mathbf{b}_{i_{|I|}} + \frac{1}{2} e_{I''} \sqcup \mathbf{a}_{i_{|I|-1} i_{|I|}}, \quad (6)$$

for $\mathbf{b}$ and $\mathbf{a}$ as in Lemma 2.2.

Proof Let $\sigma_j(Y_t)$ denote the j -th row of $\sigma(Y_t)$. Using the definition of the signature, the Stratonovich integral and the shuffle property for each $|I| \geq 0$ we can compute

$$\begin{aligned}
\langle e_I, \mathbb{Y}_t \rangle &= \int_0^t \langle e_{I'}, \mathbb{Y}_s \rangle \circ d\langle e_{i_{|I|}}, Y_s \rangle \\
&= \int_0^t \langle e_{I'}, \mathbb{Y}_s \rangle d\langle e_{i_{|I|}}, Y_s \rangle + \frac{1}{2} \int_0^t \langle e_{I''}, \mathbb{Y}_s \rangle d[\langle e_{i_{|I|-1}}, Y_s \rangle, \langle e_{i_{|I|}}, Y_s \rangle]_s \\
&= \int_0^t \langle e_{I'}, \mathbb{Y}_s \rangle \langle \mathbf{b}_{i_{|I|}}, \mathbb{Y}_s \rangle ds + \int_0^t \langle e_{I'}, \mathbb{Y}_s \rangle \sigma_{i_{|I|}}(Y_s) dW_s \\
&\quad + \frac{1}{2} \int_0^t \langle e_{I''}, \mathbb{Y}_s \rangle \langle \mathbf{a}_{i_{|I|-1}, i_{|I|}}, \mathbb{Y}_s \rangle ds \\
&= \int_0^t \langle e_{I'} \sqcup \mathbf{b}_{i_{|I|}}, \mathbb{Y}_s \rangle ds + \int_0^t \langle e_{I'}, \mathbb{Y}_s \rangle \sigma_{i_{|I|}}(Y_s) dW_s \\
&\quad + \frac{1}{2} \int_0^t \langle e_{I''} \sqcup \mathbf{a}_{i_{|I|-1}, i_{|I|}}, \mathbb{Y}_s \rangle ds \\
&= \int_0^t \langle Le_I, \mathbb{Y}_s \rangle ds + \int_0^t \langle e_{I'}, \mathbb{Y}_s \rangle \sigma_{i_{|I|}}(Y_s) dW_s.
\end{aligned}$$

Fix $|I| \leq n$. Since $|I \sqcup J| = |I| + |J|$ we get that $L(T^{(n)}(\mathbb{R}^d)) \subseteq T^{(n)}(\mathbb{R}^d)$ and thus that the corresponding drift components are linear maps in $\mathbb{Y}^n$. Similarly, fixing $|I|, |J| \leq n$ and using that $\mathbf{a}_{ij} = \sum_{|I|, |J| \leq 1} \lambda_{ij}^{IJ} e_I \sqcup e_J$ for some $\lambda_{ij}^{IJ} \in \mathbb{R}$ we get that

$$\begin{aligned}
\langle e_{I'}, \mathbb{Y}_s \rangle \sigma_{i_{|I|}}(Y_t) (\langle e_{J'}, \mathbb{Y}_s \rangle \sigma_{i_{|J|}}(Y_t))^\top &= \langle e_{I'}, \mathbb{Y}_s \rangle \langle e_{J'}, \mathbb{Y}_s \rangle \langle \mathbf{a}_{i_{|I|}j_{|J|}}, \mathbb{Y}_s \rangle \\
&= \sum_{|H_1|, |H_2| \leq 1} \lambda_{i_{|I|}j_{|J|}}^{H_1 H_2} \langle e_{I'} \sqcup e_{H_1}, \mathbb{Y}_s \rangle \langle e_{J'} \sqcup e_{H_2}, \mathbb{Y}_s \rangle.
\end{aligned}$$

This shows that the components of the corresponding diffusion matrix are polynomials of degree 2 in $\mathbb{Y}^n$. The polynomial property follows from Lemma 2.2 in [19]. $\square$

As the linear operator L maps the finite-dimensional vector space $T^{(n)}(\mathbb{R}^d)$ into itself, it can be represented by a matrix.

Definition 2.4 We call the operator L defined in (6) *dual operator corresponding to $\mathbb{Y}$* and denote by G the d_n -dimensional matrix representative of L . Explicitly, for each $|I| \leq n$ we consider coefficients $\eta_{IJ} \in \mathbb{R}$ such that

$$Le_I = \sum_{|J| \leq n} \eta_{IJ} e_J,$$

and fix a labelling injective function $\mathcal{L} : \{I : |I| \leq n\} \rightarrow \{1, \dots, d_n\}$. The matrix $G \in \mathbb{R}^{d_n \times d_n}$ is then given by

$$G_{\mathcal{L}(I), \mathcal{L}(J)} := \eta_{IJ}. \quad (7)$$

Note that using the notation of (4), for each $\mathbf{u} \in T^{(n)}(\mathbb{R}^d)$ it holds

$$\mathbf{vec}(L\mathbf{u}) = G\mathbf{vec}(\mathbf{u}).$$

The results of the following theorem have been implemented and the code is available at the repository github.com/sarasvaluto/AffPolySig presented in [13].

Theorem 2.5 *Let $(Y_t)_{t \geq 0}$ be a polynomial process satisfying (5), G be the d_n -dimensional matrix representative of the dual operator corresponding to $\mathbb{Y}$, and $(\mathcal{F}_t)_{t \geq 0}$ be the filtration generated by $(Y_t)_{t \geq 0}$. Then for each $T, t \geq 0$ and each $|I| \leq n$ it holds*

$$\mathbb{E}[\langle e_I, \mathbb{Y}_{T+t}^n \rangle | \mathcal{F}_T] = \sum_{|J| \leq n} (e^{tG^\top})_{\mathcal{L}(I), \mathcal{L}(J)} \langle e_J, \mathbb{Y}_T^n \rangle,$$

or equivalently,

$$\mathbb{E}[\mathbf{vec}(\mathbb{Y}_{T+t}^n) | \mathcal{F}_T] = e^{tG^\top} \mathbf{vec}(\mathbb{Y}_T^n),$$

where $e^{(\cdot)}$ denotes the matrix exponential.

Proof Lemma 2.3 yields that $\mathbf{vec}(\mathbb{Y}^n)$ is a polynomial process and the claim follows by Theorem 3.1 in [19] for polynomials of degree 1. $\square$

In the special case given by $Y = W$ the matrix G is nilpotent and we obtain a more explicit representation. Loosely speaking, the signature element $\langle e_I, \widehat{\mathbb{W}}_t \rangle$ has nonzero expectation only if all the indices $i_j \neq 0$ in I are grouped in blocks of even length.

Corollary 2.6 *Let W be a vector of d correlated Brownian motions with correlation matrix ρ . Consider a multi-index I admitting the representation*

$$e_I = e_0^{\otimes k_0} \otimes e_{J_1} \otimes e_0^{\otimes k_1} \otimes e_{J_2} \otimes \cdots \otimes e_0^{\otimes k_m}, \quad (8)$$

for some $J_i \in \{1, \dots, d\}^{2h_i}$ and $h_i, k_i \in \mathbb{N}_0$. Then,

$$\mathbb{E}[\langle e_I, \widehat{\mathbb{W}}_t \rangle] = \frac{t^{\sum_{i=0}^m k_i + \sum_{i=1}^m h_i}}{(\sum_{i=0}^m k_i + \sum_{i=1}^m h_i)!} \left(\frac{1}{2}\right)^{\sum_{i=1}^m h_i} \prod_{i=1}^m \rho(J_i), \quad (9)$$

where $\rho(J) := \prod_{k=1}^{|J|/2} \rho_{j_{2k-1}, j_{2k}}$. If I does not admit representation (8) then $\mathbb{E}[\langle e_I, \widehat{\mathbb{W}}_t \rangle] = 0$. Additionally, for each $s, t \geq 0$ it holds

$$\mathbb{E}[\langle e_I, \widehat{W}_{s+t} \rangle | \mathcal{F}_s] = \sum_{e_{I_1} \otimes e_{I_2} = e_I} \langle e_{I_1}, \widehat{W}_s \rangle \mathbb{E}[\langle e_{I_2}, \widehat{W}_t \rangle].$$

Note that letting I_0 be the number of 0s in I and setting $I_{>0} := |I| - I_0$ we can rewrite (9) as

$$\mathbb{E}[\langle e_I, \widehat{W}_t \rangle] = \frac{t^{I_0+I_{>0}/2}}{(I_0 + I_{>0}/2)!} \left(\frac{1}{2}\right)^{\frac{I_{>0}}{2}} \prod_{i=1}^m \rho(J_i).$$

Example 2.7 Note that in the special case where W is a d -dimensional (standard) Brownian motion we get that $\rho_{ij} = 1_{\{i=j\}}$. Corollary 2.6 then yields

$$\mathbb{E}[\langle e_I, \widehat{W}_t \rangle] = \frac{t^{I_0+I_{>0}/2}}{(I_0 + I_{>0}/2)!} \left(\frac{1}{2}\right)^{\frac{I_{>0}}{2}},$$

for each I of the form $e_I = e_{j_0}^{\otimes k_0} \otimes \cdots \otimes e_{j_m}^{\otimes k_m}$ with k_i even whenever $j_i > 0$.

In order to better understand Theorem 2.5 we propose two examples. The first concerns a vector of correlated Brownian motions and constitutes the core of the proof of Corollary 2.6. The second one leads to the formulas that will be applied later on in the chapter.

Example 2.8 Let W be a d -dimensional Brownian motion and a process $(X_t)_{t \geq 0}$ given by

$$dX_t = \sqrt{a(X_t)} dW_t, \quad X_0 = x_0,$$

for some matrix a of the form $a_{ij}(X_t) = \sigma^i \sigma^j \rho_{ij}$ and some constant $\sigma^i > 0$ and $\rho_{ij} \in [-1, 1]$. Observe that $\widehat{X}$ satisfies (5) in $d + 1$ dimensions for

$$b_j(\widehat{X}_t) = 1_{\{j=0\}} \quad \text{and} \quad a_{ij}(\widehat{X}_t) = \sigma^i \sigma^j \rho_{ij} 1_{\{i,j \neq 0\}}.$$

The corresponding $\mathbf{b}$ and $\mathbf{a}$ are given by $\mathbf{b}_j = e_0 1_{\{j=0\}}$ and $\mathbf{a}_{ij} = e_0 \sigma^i \sigma^j \rho_{ij} 1_{\{i,j \neq 0\}}$ and we thus get

$$Le_I = e_{I'} 1_{\{i_{|I|}=0\}} + \frac{1}{2} e_{I''} \sigma^{i_{|I|-1}} \sigma^{i_{|I|}} \rho_{i_{|I|-1} i_{|I|}} 1_{\{i_{|I|-1}, i_{|I|} \neq 0\}}.$$

This for instance implies that $L(e_1) = 0$, $L(e_I \otimes e_0) = e_I \lrcorner \mathbf{b}_0 + \frac{1}{2} e_{I'} \lrcorner \mathbf{a}_{i_{|I|} 0} = e_I$, and $L(e_0 \otimes e_1 \otimes e_2) = \frac{1}{2} e_0 \sigma^1 \sigma^2 \rho_{12}$. Letting $(\mathcal{F}_t)_{t \geq 0}$ be the filtration generated by $(\widehat{X}_t)_{t \geq 0}$, by Theorem 2.5 we can thus conclude that

$$\mathbb{E}[\mathbf{vec}(\widehat{X}_{T+t}^n) | \mathcal{F}_T] = e^{tG^\top} \mathbf{vec}(\widehat{X}_T^n), \quad (10)$$

or equivalently,

$$\mathbb{E}[\langle e_I, \widehat{\mathbb{X}}_{T+t}^n \rangle | \mathcal{F}_T] = \sum_{|J| \leq n} (e^{tG^\top})_{\mathcal{L}(I)\mathcal{L}(J)} \langle e_J, \widehat{\mathbb{X}}_T^n \rangle, \quad (11)$$

where G denotes the $(d+1)_n$ -dimensional matrix representative of L .

Example 2.9 In a similar setting as of the previous example let W be a d -dimensional Brownian motion and set $Z_t := (X_t, B_t)$ for an another one-dimensional Brownian motion $(B_t)_{t \geq 0}$. Suppose that

$$dX_t = \text{diag}(\kappa)(\theta - X_t)dt + \sqrt{a(X_t)}dW_t, \quad X_0 = x_0,$$

where $\kappa, \theta \in \mathbb{R}^d$ and $\text{diag}(\kappa)$ denotes a diagonal matrix consisting of the components of κ and a is as before. Observe that $\widehat{Z}$ satisfies (5) in $d+2$ dimensions for

$$b_j(\widehat{Z}_t) = 1_{\{j=0\}} + \kappa^j(\theta^j - \widehat{Z}_t^j)1_{\{j \neq 0\}} \quad \text{and} \quad a_{ij}(\widehat{Z}_t) = \sigma^i \sigma^j \rho_{ij} 1_{\{i, j \neq 0\}},$$

where $\kappa^{d+1} := 0, \sigma^{d+1} := 1$, and $\rho_{j(d+1)}$ is the correlation between X^j and B . The corresponding $\mathbf{b}$ is then given by $\mathbf{b}_j = e_\emptyset(1_{\{j=0\}} + \kappa^j(\theta^j - \widehat{Z}_0^j)1_{\{j \neq 0\}}) - e_j \kappa^j 1_{\{j \neq 0\}}$ and we thus get

$$\begin{aligned} Le_I &= e_{I'}(1_{\{|I|=0\}} + \kappa^{|I|}(\theta^{|I|} - \widehat{Z}_0^{|I|})1_{\{|I| \neq 0\}}) - (e_{I'} \sqcup e_{i_{|I|}})\kappa^{|I|}1_{\{|I| \neq 0\}} \\ &\quad + \frac{1}{2}e_{I''}\sigma^{|I|-1}\sigma^{|I|}\rho_{i_{|I|-1}i_{|I|}}1_{\{|I|-1, |I| \neq 0\}}. \end{aligned}$$

3 Linear Signature Models

Let us now introduce a framework for signature-based asset price models. We fix a time horizon $T > 0$, consider a d -dimensional continuous semimartingale $X := (X^1, \dots, X^d)$ and its matrix-valued quadratic covariation $[X]$. We suppose that X encodes all the information to represent the market asset S . Recall that for notational convenience we assume that S is one-dimensional.

Furthermore, we suppose that X has some tractability properties which are made precise below and which are for instance satisfied by a d -dimensional Brownian motion, but more generally also all polynomial diffusion processes.

3.1 Definition and First Properties

We regard the continuous semimartingale X as the *primary process*, being the main modeling building block. Its time extension is denoted by

$$\widehat{X}_t := (t, X_t).$$

We use e_0 for the component of $\widehat{X}$ corresponding to time and e_k for its component corresponding to X^k . Its signature is denoted by $\widehat{\mathbb{X}}_t$.

Throughout the section we then make the following standing assumption about diffusion matrix of X .

For all $i, j \in \{1, \dots, d\}$ it holds

$$d[X^i, X^j]_t = \sum_{|I| \leq m} a_{ij}^I \langle e_I, \widehat{\mathbb{X}}_t \rangle dt \quad (12)$$

for some $m \in \mathbb{N}$. Observe that this assumption is always satisfied when X is given by a polynomial process.

Remark 3.1 It could be of interest to consider the extension $\widehat{X}_t := (t, X_t, [X]_t)$, which includes the d^2 -dimensional process given by the quadratic covariation of the primary process X . In this case the components of the diffusion matrix do not need to be linear functions of the time extended signature, but could for instance be general path-dependent functionals. Moreover, the map $t \mapsto [X]_t$ does not need to be absolutely continuous with respect to the Lebesgue measure, hence X does not need to be an Itô-semimartingale. For ease of exposition we focus here on the simple time extension and refer to [11] for the slightly more general setup.

Our goal consists in approximating the dynamics of S with a *signature model*.

Definition 3.2 (Signature Model) A *signature model* is a stochastic process of the form

$$S_n(\ell)_t := \ell_\emptyset + \sum_{0 < |I| \leq n} \ell_I \langle e_I, \widehat{\mathbb{X}}_t \rangle, \quad (13)$$

where $n \in \mathbb{N}$ and $\ell := \{\ell_\emptyset, \ell_I : 0 < |I| \leq n\}$.

Remark 3.3 By Proposition 3.4 below the class of Sig-SDEs models considered in [37] can be embedded in our framework by choosing a properly extended one-dimensional Brownian motion as primary process.

In the following we list several important properties which make signature models a tractable framework for stochastic finance.

- For each $t \in [0, T]$, $S_n(\ell)_t$ is linear in $\widehat{\mathbb{X}}_t$. This in particular implies that having precomputed $\widehat{\mathbb{X}}$ an update of the parameters ℓ boils down to evaluating the scalar product in (13).
- The quadratic variation of processes of form (13) is again of the form (13).
- Representation (13) remains invariant under polynomial transformations.
- Itô-integrals of processes of form (13) with respect to processes of form (13) are again processes of form (13). This includes in particular the signature $\widehat{S}_n(\ell)$ of $\widehat{S}_n(\ell)_t := (t, S_n(\ell)_t)$ or expressions of the form $\int_0^\cdot S_n(\ell)_s dX_s^i$.
- The latter point implies that the expected signature of $\widehat{S}_n(\ell)$ can be expressed as

$$\mathbb{E}[e_J, \widehat{S}_n(\ell)_t] = P_J(\ell, \mathbb{E}[\widehat{\mathbb{X}}_t]),$$

for some P_J such that $P_J(\cdot, \mathbb{E}[\widehat{\mathbb{X}}_t])$ is a polynomial of degree $|J|$ and $P_J(\ell, \cdot)$ is a linear map for each ℓ (see Theorem 3.9 and Remark 3.11 for more details).

- Due to the universal approximation theorem (see e.g. [7, 15]) this yields approximations

$$\mathbb{E}[f((\widehat{S}_n^2(\ell)_t)_{t \in [0, T]})] \approx P_f(\ell, \mathbb{E}[\widehat{\mathbb{X}}_T])$$

for each map f , which is continuous in a suitable sense, where P_f is given by a finite linear combination of maps P_J as above and with $\widehat{S}^2$ we denote here the second order lift. This includes representations for

$$\mathbb{E}[\tilde{f}(\widehat{S}_n(\ell)_T)] \quad \text{and} \quad \mathbb{E}\left[\tilde{f}\left(\int_0^T \widehat{S}_n(\ell)_t dt\right)\right]$$

for maps $\tilde{f}$ being payoff functions and where the expectation is taken with respect to a risk-neutral measure.

We now prove several representation results for the signature model of form (13).

Proposition 3.4 *Fix $n \in \mathbb{N}$. Then, there is a one to one correspondence between representations of the form (13) and representations of the form*

$$\begin{aligned} S_n(\ell)_t = & \ell_\emptyset + \int_0^t \left(\ell_\emptyset^0 + \sum_{0 < |I| \leq n-1} \ell_I^0 \langle e_I, \widehat{\mathbb{X}}_s \rangle \right) ds \\ & + \sum_{k=1}^d \int_0^t \left(\ell_\emptyset^k + \sum_{0 < |I| \leq n-1} \ell_I^k \langle e_I, \widehat{\mathbb{X}}_s \rangle \right) dX_s^k, \end{aligned}$$

for $\ell := \{\ell_\emptyset, \ell_I^k : |I| \leq n-1 \text{ and } k \in \{0, \dots, d\}\}$.

Before presenting the proof of let us formulate the following technical lemma.

Lemma 3.5 *For each $|I| > 0$ it holds*

$$\int_0^t \langle e_I, \widehat{\mathbb{X}}_s \rangle ds = \langle e_I \otimes e_0, \widehat{\mathbb{X}}_t \rangle, \quad \text{and} \quad \int_0^t \langle e_I, \widehat{\mathbb{X}}_s \rangle dX_s^k = \langle \tilde{e}_I^k, \widehat{\mathbb{X}}_t \rangle,$$

where

$$\tilde{e}_I^k := e_I \otimes e_k - \sum_{|J| \leq m} \frac{a_{i_{|I|}k}^J}{2} (e_{I'} \sqcup e_J) \otimes e_0.$$

Proof The representations of $\int_0^t \langle e_I, \widehat{\mathbb{X}}_s \rangle ds$ follows by the definition of signature. We proceed with the proof of the representation of $\int_0^t \langle e_I, \widehat{\mathbb{X}}_s \rangle dX_s^k$. For $I = \emptyset$ the claim follows. By the definition of the Stratonovich integral,

$$\begin{aligned} \int_0^t \langle e_I, \widehat{\mathbb{X}}_s \rangle dX_s^k &= \int_0^t \langle e_I, \widehat{\mathbb{X}}_s \rangle \circ dX_s^k - \frac{1}{2} [\langle e_I, \widehat{\mathbb{X}} \rangle, X^k]_t \\ &= \langle e_I \otimes e_k, \widehat{\mathbb{X}}_t \rangle - \frac{1}{2} \int_0^t \langle e_{I'}, \widehat{\mathbb{X}}_s \rangle d[X^{i_{|I|}}, X^k]_s. \end{aligned}$$

The shuffle property and the definition of the signature yield then

$$\begin{aligned} \int_0^t \langle e_I, \widehat{\mathbb{X}}_s \rangle d[X^{k_1}, X^{k_2}]_s &= \int_0^t \sum_{|J| \leq m} a_{k_1 k_2}^J \langle e_I, \widehat{\mathbb{X}}_s \rangle \langle e_J, \widehat{\mathbb{X}}_s \rangle ds \\ &= \sum_{|J| \leq m} a_{k_1 k_2}^J \langle (e_I \sqcup e_J) \otimes e_0, \widehat{\mathbb{X}}_t \rangle, \end{aligned}$$

and the claim follows. $\square$

Suppose that X is a vector of correlated Brownian motions with correlation matrix ρ . Then, the transformation introduced in Lemma 3.5 reads

$$\tilde{e}_\emptyset^k = e_k \quad \text{and} \quad \tilde{e}_I^k = e_I \otimes e_k - \frac{\rho_{i_{|I|},k}}{2} \mathbf{1}_{\{i_{|I|} \neq 0\}} e_{I'} \otimes e_0,$$

for each $|I| > 0$.

We are now ready to provide the proof of Proposition 3.4.

Proof of Proposition 3.4 Let $S_n(\ell)$ be as in the statement of the proposition. By Lemma 3.5 it holds

$$\begin{aligned} S_n(\ell)_t &= \ell_\emptyset + \left(\ell_\emptyset^0 \langle e_0, \widehat{\mathbb{X}}_t \rangle + \sum_{0 < |I| \leq n-1} \ell_I^0 \langle e_I \otimes e_0, \widehat{\mathbb{X}}_t \rangle \right) \\ &\quad + \sum_{k=1}^d \left(\ell_\emptyset^k \langle \tilde{e}_\emptyset^k, \widehat{\mathbb{X}}_t \rangle + \sum_{0 < |I| \leq n-1} \ell_I^k \langle \tilde{e}_I^k, \widehat{\mathbb{X}}_t \rangle \right), \end{aligned}$$

which is of the form given by (13). Conversely, by Lemma 3.5 we also have

$$\begin{aligned}\langle e_I \otimes e_0, \widehat{\mathbb{X}}_t \rangle &= \int_0^t \langle e_I, \widehat{\mathbb{X}}_s \rangle ds, \\ \langle e_I \otimes e_k, \widehat{\mathbb{X}}_t \rangle &= \int_0^t \langle e_I, \widehat{\mathbb{X}}_s \rangle dX_s^k + \frac{1}{2} \sum_{|J| \leq m} \int_0^t \langle a_{i|I|k}^J(e_{I'} \sqcup e_J), \widehat{\mathbb{X}}_s \rangle ds,\end{aligned}$$

yielding the claim. $\square$

3.2 Absence of Arbitrage and Universality Properties

The results obtained in the representation above will enable us to establish conditions ensuring absence of arbitrage. More precisely, we shall make the assumption that the principle of “no free lunch with vanishing risk” [16] holds. This assumption is equivalent to the existence of an equivalent local martingale measure $\mathbb{Q}$ as $S_n(\ell)$ has continuous sample paths. Throughout this section, we will assume zero interest rates and that the asset has already been discounted. To formulate a precise condition for the absence of arbitrage, we rely on the following corollary, which is a direct consequence of Proposition 3.4 and Lemma 3.5.

Corollary 3.6 *Suppose that X is a local martingale. Then $S_n(\ell)$ is a local martingale if and only if it admits a representation of the form*

$$S_n(\ell)_t = \ell_\emptyset + \sum_{k=1}^D \left(\ell_\emptyset^k \langle e_k, \widehat{\mathbb{X}}_t \rangle + \sum_{0 < |I| \leq n-1} \ell_I^k \langle \tilde{e}_I^k, \widehat{\mathbb{X}}_t \rangle \right), \quad (14)$$

for some $D \in \{1, \dots, d\}$ and $\ell := \{\ell_\emptyset, \ell_I^k : 0 \leq |I| \leq n-1 \text{ and } k \in \{1, \dots, d\}\}$.

Proof Since X is a local martingale, $S_n(\ell)$ in the representation of Proposition 3.4 is a local martingale if and only if all integrals with respect to time and the quadratic variation process vanish. This means that $S_n(\ell)$ is of form

$$S_n(\ell) = \ell_\emptyset + \sum_{k=1}^d \int_0^t \left(\ell_\emptyset^k + \sum_{0 < |I| \leq n-1} \ell_I^k \langle e_I, \widehat{\mathbb{X}}_s \rangle \right) dX_s^k \quad (15)$$

and Lemma 3.5 yields the assertion. $\square$

Let us now formulate sufficient no-arbitrage conditions.

Corollary 3.7 *Suppose that there is an equivalent measure $\mathbb{Q} \sim \mathbb{P}$ such that X is a local $\mathbb{Q}$ -martingale. Then the following holds.*

(i) *The model $S_n(\ell)$ is free of arbitrage if it admits a representation as of (14).*

(ii) If $\mathbb{Q}$ is an equivalent local martingale measure for $S_n(\ell)$, then $S_n(\ell)$ is necessarily of form (14).

Proof The first assertion is a direct consequence of Corollary 3.6, as form (14) implies that $S_n(\ell)$ is a local $\mathbb{Q}$ -martingale and thus $\mathbb{Q}$ is an equivalent local martingale measure. The assumption of the second assertion implies that $S_n(\ell)$ is a local martingale under $\mathbb{Q}$, whence by Corollary 3.6 it has to be of form (14). $\square$

Remark 3.8

- (i) It is important to recognize that $S_n(\ell)$ could be free of arbitrage without being of the form in (14). This scenario arises when the primary process X is not a local martingale under any of the equivalent local martingale measures.
- (ii) Additionally, note that if $S_n(\ell)$ is of the form (14) and a local martingale under $\mathbb{Q}$, it does not necessarily imply that X is a local $\mathbb{Q}$ -martingale. This situation arises when the drift terms in (15) cancel each other out. It is worth mentioning that in the one-dimensional case ($d = 1$), such a scenario is impossible, and consequently, X is inevitably a local $\mathbb{Q}$ -martingale. Moreover, $\mathbb{Q}$ is unique and the model thus complete.

3.3 The Expected Signature of $S_n(\ell)$

For the pricing of so-called *sig-payoffs* discussed in Sect. 3.4, it is crucial to be able to compute the expected signature of $\widehat{S}_n(\ell)_t = (t, S_n(\ell)_t)$. In this section, we present formulas that link this computation to the calculation of the expected signature of $\widehat{X}_t$, which can often be explicitly computed. Specifically, when X is a Brownian motion, this is a well-known result (refer to, e.g., [18]). For the case where X is a polynomial process, the computation can be carried out using polynomial processes techniques (see Sect. 2). In a general setting, it can be computed by solving an infinite-dimensional system of linear PDEs corresponding to the Kolmogorov forward equation of the signature process (see [36]). For a comprehensive treatment of signature cumulants, i.e., the logarithm of the expected signature, we direct the readers to [21].

Theorem 3.9 Fix $n \in \mathbb{N}$, a multi-index J , and $D \in \{1, \dots, d\}$ and denote by $\widehat{S}_n(\ell)_t$ the signature of $\widehat{S}_n(\ell)_t$. Let e_0 be the component of $\widehat{S}_n(\ell)$ corresponding to time and e_1 its component corresponding to $S_n(\ell)$. Define $e(\emptyset, \ell) := \tilde{e}(\emptyset, \ell) := e_\emptyset$ and

$$e(J, \ell) = \widetilde{\omega}_{i=1}^{|J|} \left(e_0 \mathbf{1}_{\{j_i=0\}} + \left(\sum_{0 < |I| \leq n} \ell_I e_I \right) \mathbf{1}_{\{j_i=1\}} \right),$$

$$\tilde{e}(J, \ell) = \widetilde{\omega}_{i=1}^{|J|} \left(e_0 \mathbf{1}_{\{j_i=0\}} + \left(\sum_{k=1}^D \left(\ell_\emptyset^k e_k + \sum_{0 < |I| \leq n-1} \ell_I^k \tilde{e}_I^k \right) \right) \mathbf{1}_{\{j_i=1\}} \right),$$

for $|J| > 0$ with $\widetilde{\omega}$ the half-shuffle being introduced in Definition 1.1. Then the following representations hold.

- $\langle e_J, \widehat{S}_n(\ell)_t \rangle = \langle e(J, \ell), \widehat{X}_t \rangle$, if $S_n(\ell)$ is given by (13), and
- $\langle e_J, \widehat{S}_n(\ell)_t \rangle = \langle \widetilde{e}(J, \ell), \widehat{X}_t \rangle$, if $S_n(\ell)$ is given as in Corollary 3.6.

Proof We proceed by induction to prove the claim. Fix $S_n(\ell)$ as in (13). For $J = \emptyset$ the claim follows by the definition of signature. Suppose the claim holds true for each J such that $|J| = m - 1$, and fix J with $|J| \leq m$. Then

$$\begin{aligned}
 \langle e_J, \widehat{S}_n(\ell)_t \rangle &= \int_0^t \langle e_{J'}, \widehat{S}_n(\ell)_s \rangle \circ d\langle e_{j_m}, \widehat{S}_n(\ell)_s \rangle \\
 &= \int_0^t \langle e_{J'}, \widehat{S}_n(\ell)_s \rangle \circ d\langle e_0 \mathbf{1}_{\{j_m=0\}} + \left(\sum_{0 < |I| \leq n} \ell_I e_I \right) \mathbf{1}_{\{j_m=1\}}, \widehat{X}_s \rangle \\
 &= \mathbf{1}_{\{j_m=0\}} \int_0^t \langle e_{J'}, \widehat{S}_n(\ell)_s \rangle \circ d\langle e_0, \widehat{X}_s \rangle \\
 &\quad + \mathbf{1}_{\{j_m=1\}} \sum_{0 < |I| \leq n} \ell_I \int_0^t \langle e_{J'}, \widehat{S}_n(\ell)_s \rangle \circ d\langle e_I, \widehat{X}_s \rangle.
 \end{aligned}$$

The induction hypothesis and Remark 1.2 yield the first claim and the second one is analogous. $\square$

Remark 3.10 The linear combinations of multiindices given by $e(J, \ell)$ might look very abstract and we thus provide some additional details. The intuition behind its construction is as follows. For each 0 in J we are integrating with respect to time and we thus add $\widetilde{\omega}e_0$ to the current linear combination of multiindices. For each 1 in J we are instead integrating with respect to $S_n(\ell)$ and we thus need to add $\widetilde{\omega}(\sum_{0 < |I| \leq n} \ell_I e_I)$. Since integrating with respect to $S_n(\ell)$ corresponds to integrating with respect to $S_n(\ell) - S_n(\ell)_0$ the term $\ell_\emptyset e_\emptyset$ can be omitted. Choosing $J = (0, 1)$ and $J = (1, 1)$ we would for instance get

$$\begin{aligned}
 e((0, 1), \ell) &= e_0 \widetilde{\omega} \left(\sum_{0 < |I| \leq n} \ell_I e_I \right) = \sum_{0 < |I| \leq n} \ell_I e_0 \widetilde{\omega} e_I, \\
 e((1, 1), \ell) &= \left(\sum_{0 < |I| \leq n} \ell_I e_I \right) \widetilde{\omega} \left(\sum_{0 < |I| \leq n} \ell_I e_I \right) = \sum_{0 < |I_1|, |I_2| \leq n} \ell_{I_1} \ell_{I_2} e_{I_1} \widetilde{\omega} e_{I_2}.
 \end{aligned}$$

The intuition behind $\widetilde{e}(J, \ell)$ is similar.

Remark 3.11

- (i) Let $S_n(\ell)$ be as in (13) and observe that

$$e(J, \ell) = \widetilde{\omega}_{i=1}^{|J|} \sum_{0 < |I| \leq n} \left(e_I \mathbf{1}_{\{j_i=0\}} \mathbf{1}_{\{I=(0)\}} + \ell_I e_I \mathbf{1}_{\{j_i=1\}} \right).$$

Setting $c(j, I, \ell) := \mathbf{1}_{\{j=0\}} \mathbf{1}_{\{I=(0)\}} + \ell_I \mathbf{1}_{\{j=1\}}$ we thus obtain that

$$\begin{aligned} \mathbb{E}[\langle e_J, \widehat{S}_n(\ell)_t \rangle] &= \mathbb{E}[\langle e(J, \ell), \widehat{X}_t \rangle] \\ &= \sum_{|I_1|, \dots, |I_{|J|}|=1}^n \mathbb{E}[\langle \widetilde{\omega}_{i=1}^{|J|} e_{I_i}, \widehat{X}_t \rangle] \prod_{i=1}^{|J|} c(j_i, I_i, \ell). \end{aligned}$$

Although this representation may seem intricate at first glance, it is, in fact, quite convenient as the expectations $\mathbb{E}[\langle \widetilde{\omega}_{i=1}^{|J|} e_{I_i}, \widehat{X}_t \rangle]$ can be computed just once in advance. Since $c(j, I, \cdot)$ is affine, we also immediately obtain that the map

$$(\ell, \mathbb{E}[\widehat{X}_t]) \mapsto P_J(\ell, \mathbb{E}[\widehat{X}_t]) := \mathbb{E}[\langle e_J, \widehat{S}_n(\ell)_t \rangle]$$

is polynomial of degree $|J|$ in its first argument and linear in the second one.

(ii) Similarly, let $S_n(\ell)$ be as in Corollary 3.6, set $\tilde{e}_{\emptyset}^k := e_k$, and observe that

$$\tilde{e}(J, \ell) = \widetilde{\omega}_{i=1}^{|J|} \sum_{k=1}^D \sum_{0 \leq |I| \leq n-1} \left(\tilde{e}_I^k \mathbf{1}_{\{j_i=0\}} \mathbf{1}_{\{I=(0)\}} \mathbf{1}_{\{k=1\}} + \ell_I^k \tilde{e}_I^k \mathbf{1}_{\{j_i=1\}} \right).$$

Setting $\tilde{c}(j, I, k, \ell) := \mathbf{1}_{\{j=0\}} \mathbf{1}_{\{I=(0)\}} \mathbf{1}_{\{k=1\}} + \ell_I^k \mathbf{1}_{\{j=1\}}$ yields

$$\mathbb{E}[\langle e_J, \widehat{S}_n(\ell)_t \rangle] = \sum_{k_1, \dots, k_{|J|}=1}^D \sum_{|I_1|, \dots, |I_{|J|}|=0}^{n-1} \mathbb{E}[\langle \widetilde{\omega}_{i=1}^{|J|} \tilde{e}_{I_i}^{k_i}, \widehat{X}_t \rangle] \prod_{i=1}^{|J|} \tilde{c}(j_i, I_i, k_i, \ell).$$

We obtain that the map

$$(\ell, \mathbb{E}[\widehat{X}_t]) \mapsto \tilde{P}_J(\ell, \mathbb{E}[\widehat{X}_t]) := \mathbb{E}[\langle e_J, \widehat{S}_n(\ell)_t \rangle]$$

is polynomial of degree $|J|$ in its first argument and linear in the second one.

The expressions above also provide a formula of the variance of $S_n(\ell)$.

Corollary 3.12 *Let $S_n(\ell)$ be as in Corollary 3.6 and assume that it is a true martingale. Then*

$$\text{Var}(S_n(\ell)_t) = 2 \sum_{k_1, k_2=1}^D \sum_{|I_1|, |I_2|=1}^{n-1} \mathbb{E}[\langle \tilde{e}_{I_1}^{k_1} \sqcup \tilde{e}_{I_2}^{k_2}, \widehat{X}_t \rangle] \ell_{I_1} \ell_{I_2}. \quad (16)$$

Proof The martingale property guarantees that $\mathbb{E}[S_n(\ell)_t] = S_n(\ell)_0$ and hence, by the shuffle product, $\text{Var}(S_n(\ell)_t) = 2\mathbb{E}[\langle e_1 \otimes e_1, \widehat{S}_n(\ell)_t \rangle]$. By Remark 3.11 we can conclude that

$$\text{Var}(S_n(\ell)_t) = 2 \sum_{k_1, k_2=1}^D \sum_{|I_1|, |I_2|=1}^{n-1} \mathbb{E}[\langle \tilde{e}_{I_1}^{k_1} \sqcup \tilde{e}_{I_2}^{k_2}, \widehat{X}_t \rangle] \tilde{c}(1, I_1, k_1, \ell) \tilde{c}(1, I_2, k_2, \ell)$$

and the claim follows. $\square$

3.4 Pricing of Sig-Payoffs

We recall here the notion of *sig-payoffs* as introduced in [34].

Definition 3.13 (Sig-Payoff) Suppose that the price process S is given by a continuous semimartingale. A payoff $F : \Omega \rightarrow \mathbb{R}$ is said to be a sig-payoff if there exists $m \in \mathbb{N}$, and $f := \{f_\emptyset, f_J : 0 < |J| \leq m\}$, such that

$$F := f_\emptyset + \sum_{0 < |J| \leq m} f_J \langle e_J, \widehat{S}_T \rangle,$$

where $\widehat{S}$ denotes the signature of $\widehat{S}_t = (t, S_t)$.

Example 3.14 Let $K > 0$ be a strike price and $T > 0$ a maturity time. Then, Asian forwards written on S are payoffs of the form

$$\frac{1}{T} \int_0^T S_t dt - K = \frac{1}{T} \int_0^T (S_t - S_0) dt - K + S_0 = \frac{1}{T} \langle e_1 \otimes e_0, \widehat{S}_T \rangle + (K - S_0) \langle e_\emptyset, \widehat{S}_T \rangle,$$

and are thus sig-payoffs.

While standard vanilla derivatives like call and put options do not fall under the category of sig-payoffs, approximate sig-payoffs can still serve as efficient control variates in Monte Carlo pricing, as explained in Sect. 3.5.2.

In this section, we consistently denote $\widehat{S}_t$ as (t, S_t) and focus on pricing sig-payoffs when S follows a signature model, as specified in the subsequent corollary.

Corollary 3.15 *Let the dynamics of $S_n(\ell)$ under a local martingale measure $\mathbb{Q}$ be specified as in Corollary 3.6. Consider a sig-payoff*

$$F = f_\emptyset + \sum_{0 < |J| \leq m} f_J \langle e_J, \widehat{S}_n(\ell)_T \rangle.$$

Then, using the notation of Remark 3.11 we can write the corresponding price as

$$\begin{aligned}
\mathbb{E}_{\mathbb{Q}}[F] &= f_{\emptyset} + \sum_{0 < |J| \leq m} f_J \tilde{P}_J(\ell, \mathbb{E}_{\mathbb{Q}}[\widehat{\mathbb{X}}_T]) \\
&= f_{\emptyset} + \sum_{|J|=1}^m \sum_{k_1, \dots, k_{|J|=1}}^D \sum_{|I_1|, \dots, |I_{|J|}|=0}^n \mathbb{E}_{\mathbb{Q}}[\langle \widetilde{\omega}_{i=1}^{|J|} \tilde{e}_{I_i}^{k_i}, \widehat{\mathbb{X}}_T \rangle] f_J \prod_{i=1}^{|J|} \tilde{c}(j_i, I_i, k_i, \ell),
\end{aligned} \tag{17}$$

for $\tilde{c}(j, I, k, \ell) := \mathbf{1}_{\{j=0\}} \mathbf{1}_{\{I=(0)\}} \mathbf{1}_{\{k=1\}} + \ell_I^k \mathbf{1}_{\{j=1\}}$.

Proof This is a direct consequence of Theorem 3.9 and Remark 3.11. $\square$

Remark 3.16

- (i) Expression (17) admits also a second representation that turns out to be useful for coding:

$$\begin{aligned}
\mathbb{E}_{\mathbb{Q}}[F] &= f_{\emptyset} + \sum_{|J|=1}^m \sum_{k_1, \dots, k_{|J|=1}}^D \sum_{I_1, \dots, I_{|J|} \in \mathcal{I}} \mathbb{E}_{\mathbb{Q}}[\langle \widetilde{\omega}_{i=1}^{|J|} \tilde{e}_{I_i}^{k_i}, \widehat{\mathbb{X}}_T \rangle] \\
&\quad \times f(\mathbf{1}_{\{I_1 \neq I^t, k_1 \neq 0\}}, \dots, \mathbf{1}_{\{I_{|J|} \neq I^t, k_{|J|} \neq 0\}}) \prod_{i=1}^{|J|} \left(\mathbf{1}_{\{I_i = I^t, k_i = 0\}} + \ell_{I_i}^k \mathbf{1}_{\{I_i \neq I^t, k_i \neq 0\}} \right).
\end{aligned}$$

- (ii) With a similar procedure it is also possible to provide a representation of $\text{Var}_{\mathbb{Q}}(F)$. By the shuffle product we know that

$$(F - f_{\emptyset})^2 = \sum_{|J_1|, |J_2|=1}^m f_{J_1} f_{J_2} \langle e_{J_1} \sqcup e_{J_2}, \widehat{\mathbb{S}}_n(\ell)_T \rangle.$$

Setting $\tilde{P}_{I_1 \sqcup I_2} := \sum_{i=1}^K \tilde{P}_{J_i}$ for J_i and K satisfying $e_{I_1} \sqcup e_{I_2} = \sum_{i=1}^K e_{J_i}$ we thus obtain

$$\begin{aligned}
\text{Var}_{\mathbb{Q}}(F) &= \mathbb{E}_{\mathbb{Q}}[(F - f_{\emptyset})^2] - \mathbb{E}_{\mathbb{Q}}[F - f_{\emptyset}]^2 \\
&= \sum_{|J_1|, |J_2|=1}^m f_{J_1} f_{J_2} \left(\tilde{P}_{J_1 \sqcup J_2}(\ell, \mathbb{E}_{\mathbb{Q}}[\widehat{\mathbb{X}}_T]) - \tilde{P}_{J_1}(\ell, \mathbb{E}_{\mathbb{Q}}[\widehat{\mathbb{X}}_T]) \tilde{P}_{J_2}(\ell, \mathbb{E}_{\mathbb{Q}}[\widehat{\mathbb{X}}_T]) \right).
\end{aligned}$$

- (iii) From this representations we can see that the maps $P_F(f, \ell, \mathbb{E}_{\mathbb{Q}}[\widehat{\mathbb{X}}_T]) := \mathbb{E}_{\mathbb{Q}}[F]$ and $P_{\text{Var}(F)}(f, \ell, \mathbb{E}_{\mathbb{Q}}[\widehat{\mathbb{X}}_T]) := \text{Var}_{\mathbb{Q}}(F)$ inherit good properties from $\tilde{P}_J$. In particular, P_F is linear in f and polynomial of degree m in ℓ , and $P_{\text{Var}(F)}$ is quadratic in f and polynomial of degree $2m$ in ℓ . Both maps are linear in $\mathbb{E}[\widehat{\mathbb{X}}_T]$.

3.5 Calibration to Option Prices

In this section our objective is to calibrate models to call (and put) option prices, aiming to closely match their implied volatilities.

Throughout this section, we assume the existence of a pricing measure $\mathbb{Q}$ and that the primary process X is a vector of correlated $\mathbb{Q}$ -Brownian motions. The essence of this approach lies in selecting the parameters ℓ to reproduce the market prices of options on S . Given the prices of N call options $\pi^*(T_1, K_1), \dots, \pi^*(T_N, K_N)$ characterized by maturities T_i and strikes K_i , our goal is to determine these parameters for an optimal fit. Then, in spirit of [6], we translate the current calibration task into finding ℓ such that

$$\ell \in \operatorname{argmin}_{\ell} L_{\text{option}}(\ell),$$

where

$$L_{\text{options}}(\ell) = \sum_{i=1}^N \gamma_i \left(\pi^*(T_i, K_i) - \pi^{\text{model}}(\ell, T_i, K_i) \right)^2, \quad (18)$$

with γ_i Vega weights and $\pi^{\text{model}}(\ell, T_i, K_i)$ denoting the price of the option under the signature model with parameters ℓ , maturity T_i and strike K_i .

Remark 3.17 The loss function described in (18) can be adapted depending on the available data. A common approach is for instance to weigh the options by their bid-ask spreads, see for instance [6]. This reflects the relative importance of reproducing different option prices precisely. In the present work we shall employ Vega weights since bid-ask spreads are not always provided in the data and since they are well-suited to match the implied volatility surface from option prices, see also Remark 5.8 in [13].

For notational convenience, let us here briefly recall the definition of implied volatility, which serves as our criterion for goodness of fit. Notice that for liquidity reasons we only calibrate to vanilla options.

Definition 3.18 Let $\pi(T, K)$ be the price of a call option with maturity T and strike price K written on the asset S . The implied volatility of $\pi(T, K)$ is defined as the volatility $\sigma_{\text{IV}}(T, K)$ that solves the equation

$$\pi^{BS}(T, K, \sigma_{\text{IV}}(T, K)) = \pi(T, K), \quad (19)$$

where π^{BS} denotes the Black-Scholes option price. Then $(\sigma_{\text{IV}}(T, K))_{T, K}$ is called (implied) volatility surface, and $(\sigma_{\text{IV}}(T, K))_K$ is called (implied) volatility smile for each fixed maturity T .

Once the optimal parameters ℓ^* are found via (18) we then need to solve (19) numerically for $\pi(T, K) = \pi^{\text{model}}(\ell^*, T, K)$ to assess the goodness of fit. Indeed

we measure it in terms of the absolute relative error (in percentage) between the implied volatilities from the market and the model. Another goodness of fit test is to check whether the calibrated implied volatilities lie in the market's bid-ask spread (if available), see Sect. 4.2.

3.5.1 Monte Carlo Pricing

We now turn our attention to the task of computing the price $\pi^{\text{model}}(\ell, T, K)$ for a given strike $K \in \mathbb{R}$ and maturity $T > 0$ under the signature model.

Before presenting our Monte Carlo-based method, it is worth recalling that Sect. 3.4 provides a closed-form formula for computing sig-payoffs without relying on Monte Carlo techniques. Additionally, thanks to the universal approximation theorem, observe that call options payoffs can be approximated arbitrarily well by sig-payoffs, or even just by polynomials, on compact sets. The combination of these two properties forms the basis of the approach employed by [37], which involves breaking down the computation of $\pi^{\text{model}}(\ell, T, K)$ into two components: an approximation of call (put) payoffs using sig-payoffs and the pricing of the latter through the closed-form formula derived in Sect. 3.4. However, achieving an acceptable error between the original call payoff and the sig-payoff requires selecting a sig-payoff with signature terms of sufficiently high order. This approximation must hold for all sets of coefficients ℓ involved in the optimization procedure, encompassing a large compact set K .

It is important to note that while sig-payoffs may be used as control variates in Monte Carlo pricing techniques, as we will discuss in Sect. 3.5.2, we opted for a Monte Carlo approach. The linearity of the model makes this approach particularly manageable, rendering the entire calibration computationally feasible within a reasonable time while delivering highly accurate results.

For the Monte Carlo price we thus fix a number of samples $N_{MC} > 0$ and approximate $\pi^{\text{model}}(\ell, T, K)$ via

$$\pi^{\text{model}}(\ell, T, K) \approx \frac{1}{N_{MC}} \sum_{i=1}^{N_{MC}} (S_n(\ell)_T(\omega_i) - K)^+.$$

We stress again that this can be computed fast. Indeed, by the linearity of the model simulating $(S_n(\ell)_T(\omega_i))_{i=1}^{N_{MC}}$ boils down to the following steps:

- simulate $(X_t(\omega_i))_{t \in [0, T]}$, which in the current setting are just trajectories for correlated Brownian motions, for each $i \in \{1, \dots, N_{MC}\}$;
- compute $\langle e_I, \widehat{\mathbb{X}}_T(\omega_i) \rangle$ for all $i = 1, \dots, N_{MC}$ and for all multi-indices I such that $|I| \leq n$;
- take linear combinations to compute $\langle \tilde{e}_I, \widehat{\mathbb{X}}_T(\omega_i) \rangle$ for all $i = 1, \dots, N_{MC}$ and for all multi-indices $|I| \leq n - 1$ as described in Lemma 3.5;
- retrieve $(S_n(\ell)_T(\omega_i))_{i=1}^{N_{MC}}$ via (14).

It is worth noting that the parameters ℓ come into play only in the final step, enabling the precomputation and storage of all other quantities. This contrasts with other models where the calibration parameters are involved at each time point during the simulation steps, such as in an Euler scheme for classical Markovian models or more complex schemes employed in rough volatility models.

3.5.2 Variance Reduction with Sig-Payoffs

Even though Monte Carlo pricing is fast since all essential quantities can be precomputed as explained above, we here discuss variance reduction techniques (see e.g. [26]) that can speed up the procedure even further. The idea is to introduce a control variate, i.e., a random variable Φ^{cv} such that:

$$\mathbb{E}_{\mathbb{Q}}[\Phi^{cv}] = 0, \quad \text{Var}((S_n(\ell)_T - K)^+ - \Phi^{cv}) < \text{Var}((S_n(\ell)_T - K)^+).$$

An example of control variates used for pricing and calibrating neural SDE models can be found in [10, 25], where Φ^{cv} is constructed from the Delta hedge. A possible other choice of control variates for signature models are sig-payoffs. Indeed, one can use the pricing formula derived in Sect. 3.4 to define:

$$\Phi^{cv}(\ell, T, K) := f_{\emptyset} + \sum_{0 < |J| \leq m} f_J \langle e_J, \widehat{S}_n(\ell)_T \rangle - \widetilde{P}_f(\ell, \mathbb{E}_{\mathbb{Q}}[\widehat{X}_T]),$$

for some fixed $m > 0$ where

$$(S_n(\ell)_T - K)^+ \approx f_{\emptyset} + \sum_{0 < |J| \leq m} f_J \langle e_J, \widehat{S}_n(\ell)_T \rangle,$$

for a wide range of ℓ and with high probability. This can be done by performing a linear regression to obtain the coefficients $f = (f_J)_{|J| \leq m}$. Alternatively, a polynomial approximation of the payoff's function can also be employed.

The properties of Φ^{cv} then guarantee the accuracy of the approximation

$$\pi^{\text{model}}(\ell, T, K) \approx \frac{1}{N_{MC}} \left(\sum_{i=1}^{N_{MC}} (S_n(\ell)_T(\omega_i) - K)^+ - \Phi^{cv}(\ell, T, K)(\omega_i) \right),$$

already for smaller values of N_{MC} . In the following remark we report a numerical experiment in this regard.

Example 3.19 Fix $n, d = 2$ and consider the parameters $\ell^* \in \mathbb{R}^{13}$ calibrated to the first smile of the market data shown in Fig. 4 (left). Consider as example a call option with maturity $T = 30$ days and strike $K = 85\%$ of the spot price. Let p be a polynomial of order $m = 4$ approximating the function $f : x \mapsto (x - K)^+$ on the compact $[70\%S_0, 120\%S_0]$. Recall that $\mathbb{E}_{\mathbb{Q}}[p(S_n(\ell)_T)]$

can be computed analytically as polynomials are sig-payoffs. Using $\Phi^{cv}(\ell) := p(S_n(\ell)_T) - \mathbb{E}_{\mathbb{Q}}[p(S_n(\ell)_T)]$ as control variate we can reduce the sample variance of the Monte Carlo estimator from approximately $3.89 \cdot 10^{-5}$ to $6.5 \cdot 10^{-7}$. Observe that $p(S_n(\ell)_T)$ here coincides with the sig-payoff without time-augmentation i.e.,

$$p(S_n(\ell)_T) = \sum_{|I| \leq 4} f_I \langle e_I, \mathbb{S}_n(\ell)_T \rangle.$$

3.5.3 Model Performance

In the following we discuss the problem of minimizing the functional (18) using the Monte Carlo method as described in Sect. 3.5.1 to compute the model prices. We consider the model described in Corollary 3.6 for two correlated Brownian motions B and W with correlation coefficient $\rho = -0.5$, and $D = 1$.

As a first example we consider synthetic data, where the implied volatility surface to fit is generated by a Heston model whose dynamics are given by

$$\begin{aligned} dS_t &= \mu S_t dt + S_t \sqrt{V_t} dB_t^{\mathbb{P}} \\ dV_t &= \kappa(\theta - V_t)dt + \sigma \sqrt{V_t} dW_t^{\mathbb{P}}, \end{aligned} \quad (20)$$

where in both cases $d[B^{\mathbb{P}}, W^{\mathbb{P}}]_t = \rho dt$ where $\rho \in [-1, 1]$.

We consider 7 maturities $(T_k)_{k=1}^7$ ranging from 30 days to 2 years and 13 strikes $(K_j)_{j=1}^{13}$ ranging from 80% to 120% of the spot price. The truncation parameter is fixed to $n = 3$ and the number of Monte Carlo samples to $N_{MC} = 10^6$. The results for the following two sets of parameters under a risk neutral measure

κ	θ	σ	ρ	V_0
0.2	0.3	0.5	-0.5	0.08

are displayed in the first and in the second row of Fig. 1, respectively.

We stress that these calibrations to Heston generated implied volatility surfaces can take between 4 and 15 minutes on a standard laptop. We consider now implied volatility data as of 17/03/2021 for call options written on the S&P 500 index. Our dataset provided by Bloomberg consists of 7 maturities $(T_k)_{k=1}^7$, ranging from 30 days to 2 years, and 9 strike prices $(K_j)_{j=1}^9$ for each maturity which vary between 80% and 120% of the spot price. Again, the truncation parameter is fixed to $n = 3$ and the Monte Carlo's parameter to $N_{MC} = 10^6$. The results are displayed in Fig. 2.

Figure 2 indicates that the volatility smiles for short maturities have not been captured adequately. Specifically, as with many continuous models, the challenge lies in fitting the shortest maturities, primarily due to the high at-the-money skew in the market (refer to Chapter 3 and Chapter 7 of [22]).

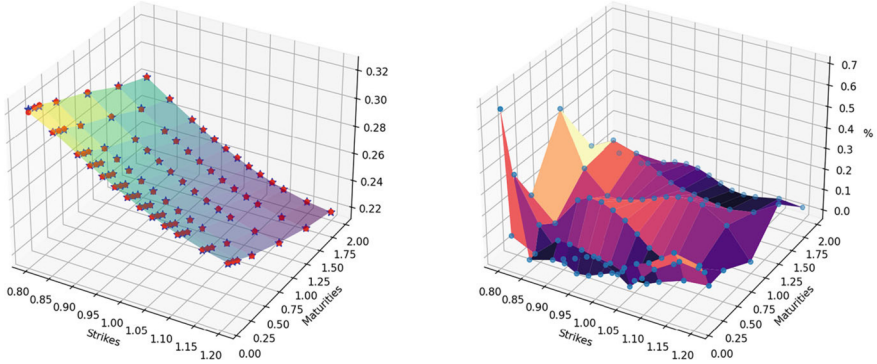


Fig. 1 On the left: blue stars correspond to the implied volatilities of the Heston models, red dots denote the calibrated implied volatilities of $S_n(\ell)$ with $n = 3$ (13 estimated parameters). On the right: absolute relative errors between the two surfaces are expressed in percentages

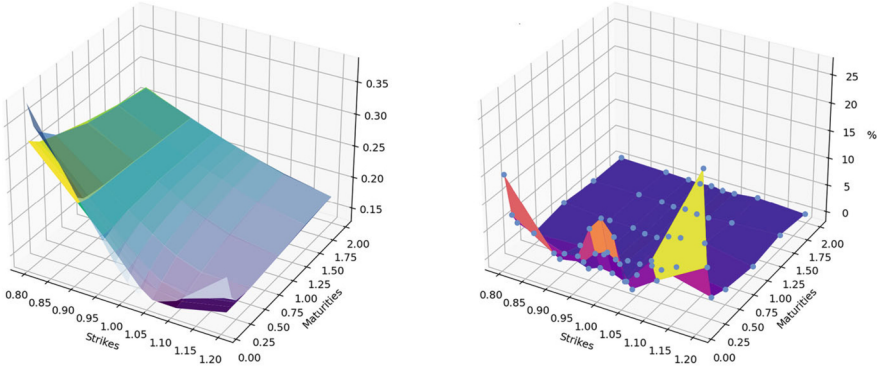


Fig. 2 On the left: the upper surface represents the implied volatility of the S&P 500 index as of 17-03-2021, the lower one is the calibrated implied volatility of $S_n(\ell^*)$ with $n = 3$ (13 parameters). On the right: absolute relative error between the two surfaces in percentages

To assess the model's ability to replicate short maturity smiles, we conduct calibrations using different loss functions that penalize outliers more severely. In the initial calibration procedure described earlier, we denote by w_i the absolute error between the target implied volatility and the approximated one for maturity T_i and strike K_i . Subsequently, inspired by generative-adversarial distances as considered in, e.g., [10], we define a new loss function

$$L_{p,\alpha}(\ell) = \sum_{i=1}^N (\gamma_i + \alpha w_i) |\pi^*(T_i, K_i) - \pi^{\text{model}}(\ell, T_i, K_i)|^p, \quad (21)$$

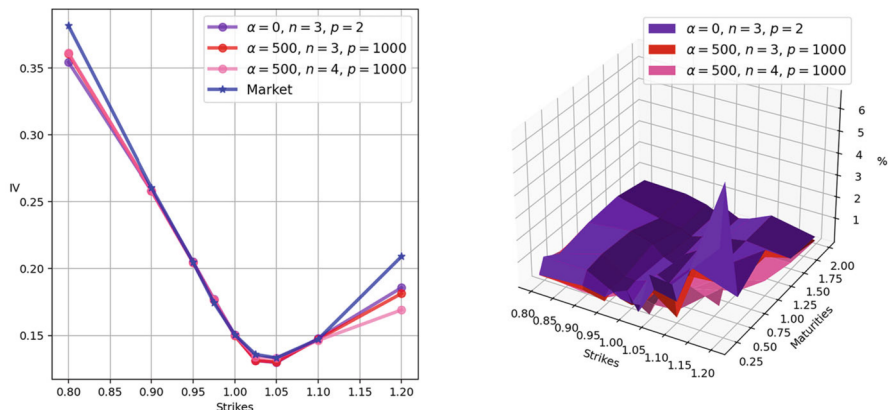


Fig. 3 Comparison between the calibrated implied volatility smiles for different parameters and the S&P 500 index as of 17-03-2021 smile (in blue) at maturity $T_1 = 30$ days (on the left) and for maturities ranging from 60 days to 2 years (on the right). Calibration has been performed using (21)

depending on parameters p and α that need to be chosen. By taking high values for p and α we can approximate the sup-distance between the two price surfaces, i.e.,

$$L_\infty(\ell) := \sup_{i=1, \dots, N} |\pi^*(T_i, K_i) - \pi^{\text{model}}(\ell, T_i, K_i)|,$$

without compromising differentiability with respect to ℓ . The result for different choices of the parameters α and p but also the truncation level n is displayed in Fig. 3. As can be guessed from the figure, although the maximal absolute relative error for maturities larger than 60 days is (almost) acceptable (5.6% for $n = 3$, $p = 1000$, and $\alpha = 500$ and 2.3 % for $n = 4$ and $p = 1000$, and $\alpha = 500$), the absolute relative error for the shortest maturity and the far in and out of the money strikes are still above 12 % and 18 %, respectively. Observe that the performance for $n = 4$ is the best for every maturity larger than 60 days as well as for the at-the-money region of the shortest maturity.

In a final experiment, we conduct two separate calibrations: one to the shortest maturity alone and another to every other maturity combined. The first calibration is performed for $T_1 = 30$ days, and the second calibration encompasses 6 maturities ranging from 60 days to 2 years. In both cases, we consider 9 strikes ranging from 80% to 120% of the spot price. For the first calibration, we fix the parameters at $n = 2$, $p = 2$, and $\alpha = 0$, while for the subsequent maturities, the parameters are set to $n = 4$, $p = 300$, and $\alpha = 500$. The results are displayed in Fig. 4. Notably, the fit for the first maturity is remarkably accurate, as well as for the entire implied volatility surface. The computational time required for the calibration of the first smile is approximately 2 minutes, while the calibration to the remaining surface might take longer, mainly due to the higher value of n and the calibration to all maturities.

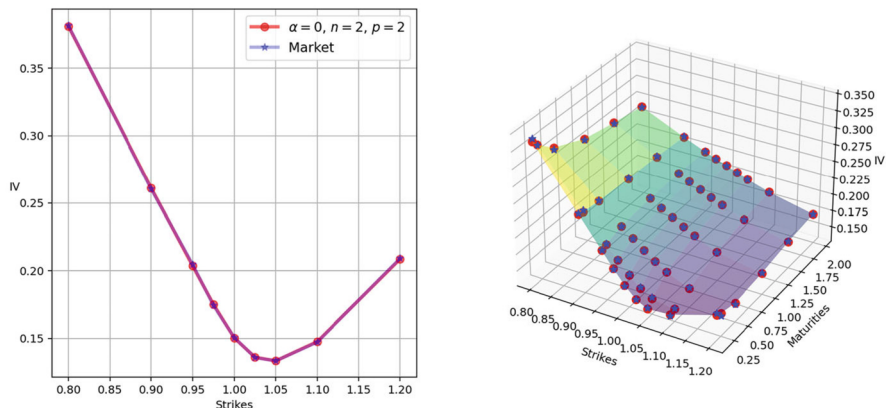


Fig. 4 Comparison between the calibrated implied volatility smiles and the SPX-500 Index as of 17-03-2021 smile (in blue) at maturity $T_1 = 30$ days (on the left) and for maturities ranging from 60 days to 2 years (on the right). Two different calibrations have been performed using (21)

These results suggest that introducing maturity-dependent parameters and performing a slice-wise calibration to the individual smiles as for instance in [10] or [25] can be of interest to obtain both, an excellent accuracy and a low computational time. For details in this regard and the out-of-sample performance see [11].

4 A Signature Model for Index Options and Volatility Derivatives

This section is dedicated to define the second model $(S_t)_{t \geq 0}$ for the S&P 500 index in detail. Under a risk-neutral probability measure $\mathbb{Q}$ the model's dynamics are given by:

$$dS_t = S_t \sigma_t^S dB_t. \quad (22)$$

Here, $S_0 \in \mathbb{R}^+$, $\sigma^S = (\sigma_t^S)_{t \geq 0}$ is the volatility process, and $B = (B_t)_{t \geq 0}$ is a one-dimensional Brownian motion that is correlated with the volatility process. Note that the instantaneous variance is given by $V_t := (\sigma_t^S)^2$ for every $t \geq 0$. Choosing a functional form of the volatility process is the crucial modelling choice. We set

$$\sigma_t^S(\ell) := \ell_\emptyset + \sum_{0 < |I| \leq n} \ell_I \langle e_I, \widehat{\mathbb{X}}_t \rangle, \quad (23)$$

i.e. the volatility process is determined by a linear function of the (time-extended) signature of a primary process X . Moreover, we assume that X is a polynomial

process (recall Definition 2.1) and that the model parameters are $\ell := \{\ell_I \in \mathbb{R} : |I| \leq n\} \in \mathbb{R}^{(d+1)_n}$.

For later convenience, we now introduce the process $(Z_t)_{t \geq 0}$ where $Z_t = (X_t, B_t)$ and denote by $(\tilde{Z}_t)_{t \geq 0}$ its time-extended signature. The correlation of the components of $(Z_t)_{t \geq 0}$ is given by $\rho = (\rho_{ij})_{i,j=1,\dots,d+1}$, i.e.,

$$\rho_{ij} = \frac{[Z^i, Z^j]}{\sqrt{[Z^i]} \sqrt{[Z^j]}} \in [-1, 1],$$

Recall that we use $[\cdot, \cdot]$ for the quadratic variation. We will often write $(\sigma_t^S)_{t \geq 0} = (\sigma_t^S(\ell))_{t \geq 0}$ as in (22) to keep the notation light and only mention the dependence on ℓ when it is explicitly needed.

Remark 4.1 (Interest Rates and Dividends) To calibrate our model to option prices we will both need the discounted, dividend-adjusted and undiscounted, unadjusted prices. This is due to the fact that the VIX is defined via the discounted, dividend-adjusted prices by the CBOE but the claims on the S&P500 itself are written on the undiscounted, unadjusted prices. Recall that $(S_t)_{t \geq 0}$ given by (22) is the discounted, dividend-adjusted price process. Including an interest rate r and dividend q , the corresponding undiscounted, unadjusted price process is given by

$$d\tilde{S}_t = (r - q)\tilde{S}_t dt + \tilde{S}_t \sigma_t^S(\ell) dB_t.$$

The price of a call option with time to maturity $T > 0$ and strike price $K \in \mathbb{R}$ written on the S&P 500 index, is therefore

$$C(T, K) = \mathbb{E} \left[e^{-rT} (\tilde{S}_T(\ell) - K)^+ \right] = \mathbb{E} [e^{-rT} (e^{(r-q)T} S_T(\ell) - K)^+],$$

where we used that $\tilde{S}_t(\ell) = e^{(r-q)t} S_t(\ell)$.

Remark 4.2 Recall our assumption that $\hat{X}$ is a polynomial process. This is important since then $\hat{X}^n$ is a finite-dimensional polynomial process in sense of [19] and [9] and therefore the expected signature of $\hat{X}$ and conditional expected signature can be found by solving a finite-dimensional ODE, more specifically, it is given by a finite-dimensional matrix exponential as described in Sect. 2.

This assumption still leaves us with a broad class of admissible primary processes. Some prominent examples are correlated Brownian motions, Cox-Ingersoll-Ross (CIR) processes, geometric Brownian motions, Jacobi processes, OU processes, and all continuous affine processes.

Remark 4.3 Several stochastic volatility models are encapsulated in our modelling framework (23). Let us elaborate on the following:

- The Stein-Stein model, as introduced in [40], is obtained if we choose a one-dimensional OU process as our primary process $(X_t)_{t \geq 0}$ and set $n = 1$, $\ell_\emptyset = \ell_{(0)} = 0$ and $\ell_{(1)} \neq 0$.

- The SABR model, as introduced initially in [31] with $\beta = 1$. To this end we choose $(X_t)_{t \geq 0}$ to be a 1-dimensional geometric Brownian motion without drift and again let $n = 1$, with $\ell_\emptyset = \ell_{(0)} = 0$ and $\ell_{(1)} \neq 0$.
- Let $(X_t)_{t \geq 0}$ be a one-dimensional OU process with $n = 5$, $\ell_\emptyset, \ell_{(1)}, \ell_{(1,1,1)}, \ell_{(1,1,1,1,1)}$ non-zero and $\ell_I = 0$ otherwise. This results in the model described in [1], except for the fact that therein a deterministic input curve is added. Moreover, we can embed the entire set of Gaussian polynomial volatility models introduced in [1], if we allow X not to be a semimartingale and do not add a time-augmentation.

In the next section we discuss the pricing of VIX options with (23) as well as the nature of the log-price process.

4.1 Explicit Formulas for the VIX and the Log-Price

The CBOE Volatility Index (VIX) measures the market's expected volatility of the S&P 500 index. More specifically, its current value corresponds to the expected annualized change in the S&P 500 index over the 30 days ahead. Indeed, the index value is computed by

$$\text{VIX}_T = \sqrt{\mathbb{E} \left[-\frac{2}{\Delta} \log \left(\frac{S_{T+\Delta}}{S_T} \right) \middle| \mathcal{F}_T \right]}, \quad (24)$$

where $\Delta = 30$ days, S_T denotes the price process and $\mathcal{F}_T$ the filtration at time $T > 0$. Note that for *any* stochastic volatility model of the form

$$dS_t = S_t \sqrt{V_t} dB_t,$$

where $(B_t)_{t \geq 0}$ is a Brownian motion and $(V_t)_{t \geq 0}$ satisfies

$$\mathbb{E} \left[\int_0^T V_s ds \right] < \infty, \quad (25)$$

the VIX index at $T > 0$ is given by

$$\text{VIX}_T = \sqrt{\frac{1}{\Delta} \mathbb{E} \left[\int_T^{T+\Delta} V_t dt \middle| \mathcal{F}_T \right]}, \quad (26)$$

see e.g., [35]. Although there are of course put and call options written on the VIX, we from now on consider without loss of generality only call options and will simply refer to them as *VIX options*.

We shall now derive an analytical expression for the VIX index value (24) under our model, i.e. if S is defined as in (22) and (23) and for a polynomial process X . More precisely, we state in Theorem 4.4 that the value of the VIX is given by the square-root of a quadratic function in the parameters ℓ . This is achieved via polynomial technology and the computation of a matrix exponential, as we showed in Sect. 2.

Theorem 4.4 *Let the price process $S = (S_t)_{t \geq 0}$ be given by*

$$dS_t = S_t \sigma_t^S(\ell) dB_t,$$

with volatility process $\sigma^S(\ell) = (\sigma_t^S(\ell))_{t \geq 0}$ and $B = (B_t)_{t \geq 0}$ denoting a one-dimensional Brownian motion. Recall our modeling assumption, namely that $\sigma^S(\ell)$ and X satisfy (23) and note that under this (25) is fulfilled. As introduced in Sect. 2 we fix an injective labeling function $\mathcal{L} : \{I : |I| \leq n\} \rightarrow \{1, \dots, (d+1)_{2n+1}\}$ and denote by G be the $(d+1)_{(2n+1)}$ -dimensional matrix representative of the dual operator corresponding to $\widehat{\mathbb{X}}$. Then, the following expression for the VIX at time $T > 0$ holds

$$\text{VIX}_T(\ell) = \sqrt{\frac{1}{\Delta} \ell^\top Q(T, \Delta) \ell}, \quad (27)$$

with

$$Q_{\mathcal{L}(I)\mathcal{L}(J)}(T, \Delta) = \mathbf{vec}((e_I \sqcup e_J) \otimes e_0)^\top (e^{\Delta G^\top} - \text{Id}) \mathbf{vec}(\widehat{\mathbb{X}}_T^{2n+1}), \quad (28)$$

and $\text{Id} \in \mathbb{R}^{(d+1)_{2n+1} \times (d+1)_{2n+1}}$ denoting the identity matrix. Note that Q is positive semidefinite and symmetric and hence allows for a Cholesky decomposition.

Proof Note that $V_t(\ell) = (\sigma^S(\ell))^2$ can be expressed as

$$V_t(\ell) = \left(\sum_{|I| \leq n} \ell_I \langle e_I, \widehat{\mathbb{X}}_t \rangle \right)^2 = \sum_{|I|, |J| \leq n} \ell_I \ell_J \langle e_I \sqcup e_J, \widehat{\mathbb{X}}_t \rangle,$$

by the shuffle-property. Moreover, recall that continuous polynomials processes have finite moments of every degree and hence by Remark 1.2 it follows that (25) holds. Plugging-in the above equation for $V_t(\ell)$ the value of the VIX is for each $T > 0$ given by

$$\text{VIX}_T^2(\ell) = \frac{1}{\Delta} \sum_{|I|, |J| \leq n} \ell_I \ell_J \mathbb{E} \left[\int_T^{T+\Delta} \langle e_I \sqcup e_J, \widehat{\mathbb{X}}_t \rangle dt \middle| \mathcal{F}_T \right] = \frac{1}{\Delta} \ell^\top Q(T, \Delta) \ell,$$

where for each $T > 0$ the matrix Q is given by

$$\begin{aligned}
Q_{\mathcal{L}(I)\mathcal{L}(J)}(T, \Delta) &:= \mathbb{E} \left[\int_T^{T+\Delta} \langle e_I \sqcup e_J, \widehat{\mathbb{X}}_t \rangle dt | \mathcal{F}_T \right] \\
&= \mathbb{E} \left[\int_0^{T+\Delta} \langle e_I \sqcup e_J, \widehat{\mathbb{X}}_t \rangle dt - \int_0^T \langle e_I \sqcup e_J, \widehat{\mathbb{X}}_t \rangle dt | \mathcal{F}_T \right] \\
&= \mathbb{E} [\langle (e_I \sqcup e_J) \otimes e_0, \widehat{\mathbb{X}}_{T+\Delta} \rangle - \langle (e_I \sqcup e_J) \otimes e_0, \widehat{\mathbb{X}}_T \rangle | \mathcal{F}_T] \\
&= \mathbb{E} [\langle (e_I \sqcup e_J) \otimes e_0, \widehat{\mathbb{X}}_{T+\Delta} \rangle | \mathcal{F}_T] - \langle (e_I \sqcup e_J) \otimes e_0, \widehat{\mathbb{X}}_T \rangle.
\end{aligned}$$

By Theorem 2.5 we can also express Q as

$$\begin{aligned}
Q_{\mathcal{L}(I)\mathcal{L}(J)}(T, \Delta) &= \mathbf{vec}((e_I \sqcup e_J) \otimes e_0)^\top e^{\Delta G^\top} \mathbf{vec}(\widehat{\mathbb{X}}_T^{2n+1}) \\
&\quad - \mathbf{vec}((e_I \sqcup e_J) \otimes e_0)^\top \mathbf{vec}(\widehat{\mathbb{X}}_T^{2n+1}) \\
&= \mathbf{vec}((e_I \sqcup e_J) \otimes e_0)^\top (e^{\Delta G^\top} - \text{Id}) \mathbf{vec}(\widehat{\mathbb{X}}_T^{2n+1}),
\end{aligned}$$

which proves our claim. The fact that Q is positive semidefinite and symmetric follows from the shuffle property. Under these properties, it is well-known that a Cholesky decomposition exists, meaning that there is an upper triangular matrix $U_T \in \mathbb{R}^{(d+1)_n \times (d+1)_n}$ such that

$$\text{VIX}_T(\ell) = \sqrt{\frac{1}{\Delta} \ell^\top U_T U_T^\top \ell} = \frac{1}{\sqrt{\Delta}} \sqrt{(U_T^\top \ell)^2} = \frac{1}{\sqrt{\Delta}} \|U_T^\top \ell\|.$$

□

Moreover, an explicit expression of the log price in terms of a linear function of the signature of $\widehat{Z}$ can be obtained, which we present in the next proposition.

Proposition 4.5 *Let S be given by (22) with $S_0 = 1$, as well as σ^S and X satisfy (23). Under these assumptions the log-price at time $t \geq 0$ is given by,*

$$\log(S_t(\ell)) = -\frac{1}{2} \ell^\top Q^0(t) \ell + \sum_{|I| \leq n} \ell_I \langle \tilde{e}_I^B, \widehat{\mathbb{Z}}_t \rangle, \quad (29)$$

with

$$\tilde{e}_\emptyset^B := e_{d+1}, \quad \tilde{e}_I^B := e_I \otimes e_{d+1} - \sum_{|J| \leq m} \frac{a_{i_{|I|}(d+1)}^J}{2} (e_{I'} \sqcup e_J) \otimes e_0,$$

for each $|I| > 0$, and the matrix $Q^0(t) \in \mathbb{R}^{(d+1)_n \times (d+1)_n}$ has components

$$Q_{\mathcal{L}(I)\mathcal{L}(J)}^0(t) = \langle (e_I \sqcup e_J) \otimes e_0, \widehat{\mathbb{X}}_t \rangle,$$

for an arbitrary but fixed labeling function $\mathcal{L} : \{I : |I| \leq n\} \rightarrow \{1, \dots, (d+1)_n\}$.

Proof Using Itô's lemma and the form of S , σ^S and X under our model, we obtain

$$\begin{aligned}
 \log(S_t(\ell)) &= -\frac{1}{2} \int_0^t V_s(\ell) ds + \int_0^t \sigma_s^S(\ell) dB_s \\
 &= -\frac{1}{2} \sum_{|I|, |J| \leq n} \ell_I \ell_J \int_0^t \langle e_I \sqcup e_J, \widehat{\mathbb{X}}_s \rangle ds + \sum_{|I| \leq n} \ell_I \int_0^t \langle e_I, \widehat{\mathbb{X}}_s \rangle dB_s \\
 &= -\frac{1}{2} \sum_{|I|, |J| \leq n} \ell_I \ell_J \langle (e_I \sqcup e_J) \otimes e_0, \widehat{\mathbb{X}}_t \rangle + \sum_{|I| \leq n} \ell_I \langle \tilde{e}_I^B, \widehat{\mathbb{Z}}_t \rangle \\
 &= -\frac{1}{2} \ell^\top Q^0(t) \ell + \sum_{|I| \leq n} \ell_I \langle \tilde{e}_I^B, \widehat{\mathbb{Z}}_t \rangle,
 \end{aligned}$$

where we used that $\int_0^t \langle e_I, \widehat{\mathbb{X}}_s \rangle dB_s = \langle \tilde{e}_I^B, \widehat{\mathbb{Z}}_t \rangle$ (Lemma 3.5) in the second to last equality. $\square$

4.2 Joint Calibration of SPX and VIX Options

We point out again that we only work with call options in the following, however this choice was made merely for simplicity and it is straightforward to include any other liquid options. The time-to-maturities of options available on the market, as well as strikes-prices available on the market may differ for VIX and SPX. Accordingly, we denote by $\mathcal{T}^{\text{SPX}}$, $\mathcal{T}^{\text{VIX}}$ the set of maturities and by $\mathcal{K}^{\text{SPX}}$, $\mathcal{K}^{\text{VIX}}$ the set of strikes for SPX and VIX options respectively.

Proposition 4.5 implies that the SPX call option payoff is given by

$$e^{-rT} (\tilde{S}_T(\ell) - K)^+ = e^{-rT} \left(\exp \left\{ (r-q)T - \frac{1}{2} \ell^\top Q^0(t) \ell + \sum_{|I| \leq n} \ell_I \langle \tilde{e}_I^B, \widehat{\mathbb{Z}}_T \rangle \right\} - K \right)^+,$$

for a given maturity $T > 0$ and a strike price $K > 0$. Recall from Remark 4.1 that $\tilde{S}$ stands for the undiscounted, unadjusted price process, r is the interest rate and q the dividend. Similarly, according to Theorem 4.4, the pay-off for the VIX reads

$$\begin{aligned}
 e^{-rT} (\text{VIX}_T(\ell) - K)^+ &= e^{-rT} \left(\sqrt{\frac{1}{\Delta} \ell^\top Q(T, \Delta) \ell} - K \right)^+ \\
 &= e^{-rT} \left(\frac{1}{\sqrt{\Delta}} \|U_T^\top \ell\| - K \right)^+,
 \end{aligned}$$

where we applied the Cholesky decomposition to $Q(T, \Delta) = U_T U_T^\top$. To evaluate the expectations in the pricing formula for the above options as well as the VIX's futures, we apply a Monte-Carlo procedure. However, we want to point out that we do not need an additional Monte-Carlo simulation to evaluate the conditional expectation in the definition of the VIX. This is a significant computational benefit of our model thanks to the polynomial technology.

The loss function to be minimized in the joint calibration is simply a convex combination of the individual loss functions of VIX and SPX respectively:

$$L_{\text{joint}}(\ell, \lambda) := \lambda L_{\text{SPX}}(\ell) + (1 - \lambda) L_{\text{VIX}}(\ell), \quad (30)$$

for $\lambda \in (0, 1)$. In particular, for some loss function $\mathcal{L}^\beta$ to be specified below,

- the VIX' loss function is given by

$$L_{\text{VIX}}(\ell) := \sum_{\substack{T \in \mathcal{T}^{\text{VIX}} \\ K \in \mathcal{K}^{\text{VIX}}}} \mathcal{L}^\beta \left(\pi_{\text{VIX}}^{\text{model}}(\ell, T, K), \pi_{\text{VIX}}^{b,a}(T, K), \sigma_{\text{VIX}}^{b,a}(T, K), F_{\text{VIX}}^{\text{model}}(\ell, T), F_{\text{VIX}}^{\text{mkt}}(T) \right),$$

where $\pi_{\text{VIX}}^{\text{model}}$ and $F_{\text{VIX}}^{\text{model}}$ are the Monte Carlo option- and futures-prices under our model (i.e. for $\text{VIX}_T(\ell, \omega_i)$ defined as in (27));

- the SPX' loss function is given by

$$L_{\text{SPX}}(\ell) := \sum_{T \in \mathcal{T}^{\text{SPX}}, K \in \mathcal{K}^{\text{SPX}}} \mathcal{L}^\beta(\pi_{\text{SPX}}^{\text{model}}(\ell, T, K), \pi_{\text{SPX}}^{b,a}(T, K), \sigma_{\text{SPX}}^{b,a}(T, K)).$$

Note that in the case of the SPX no future prices need to be calibrated (because the index value can directly be used as spot price) and we therefore do not add them as inputs to $\mathcal{L}^\beta$ by slight abuse of notation.

For some $\beta \in (0, 1)$, we define $\mathcal{L}^\beta$ to be the loss function

$$\mathcal{L}^\beta(\pi, \pi^{mkt,b,a}, \sigma^{mkt,b,a}, F, F^{mkt}) = \left(\frac{(\beta \tilde{1}_{\{\pi \notin \pi^{mkt,b,a}\}} + (1-\beta)) |\pi - (\pi^{mkt,a} + \pi^{mkt,b})/2| + |\delta^{mkt} e^{-rT} (F - F^{mkt})|}{\upsilon^{mkt} (\sigma^{mkt,a} - \sigma^{mkt,b})} \right)^2, \quad (31)$$

with

- $\upsilon^{mkt}, \delta^{mkt}$ denoting the Vega and Delta of the option under the Black-Scholes model (but note that they depend on both maturity and strike price);
- $\pi^{mkt,b,a} = [\pi^{mkt,b}, \pi^{mkt,a}]$, $\sigma^{mkt,b,a} = [\sigma^{mkt,b}, \sigma^{mkt,a}]$, with $\pi^{mkt,b}, \pi^{mkt,a}, \sigma^{mkt,b}$ and $\sigma^{mkt,a}$ denoting the market bid and ask prices and implied volatilities respectively.

- F and F^{mkt} standing for the models and markets futures respectively (with maturity T);
- $\tilde{I}_{\{x \notin [y^b, y^a]\}} := s(y^b - x) + s(x - y^a)$ for $s(x) := \frac{1}{2} \tanh(100x) + \frac{1}{2}$ a smooth approximation of the indicator function, which penalizes implied volatilities that lie out of the bid-ask spread.

4.3 Numerical Results

Before presenting the numerical results concerning the joint calibration problem within our framework, it is worth to mention that two main approaches in the literature are taken into account when choosing the maturities to fit.

1. One approach is to choose $T_1^{\text{SPX}} = T_1^{\text{VIX}}$ the first maturity to be the same for VIX and SPX and then for higher maturities to set $j \geq 2$, $T_j^{\text{SPX}} = T_{j-1}^{\text{VIX}} + \Delta$, see for instance [24, 27, 29, 30]. Sometimes, one may also choose the first maturity to differ by a few days between SPX and VIX, if the same maturity is not available for both on the market.
2. Another approach consists of taking the same set of maturities for SPX and VIX $\mathcal{T}^{\text{SPX}} = \mathcal{T}^{\text{VIX}}$ (or close together) as it was done for example by [4, 23, 39].

In the following we will adhere to the first approach and refer the reader to [13] for results regarding the second approach.

The trading day we consider for our calibration is 02/06/2021, the same as was used in [24, 30] and we use call options for both VIX and SPX. Maturities are reported in the following tables with the corresponding range of strikes (in percentage) with respect to the spot and the market's futures prices.

In the table below, we report the maturities used in the calibrations (where $T = 1 \equiv 365.25$ days) and report the corresponding moneyness range (i.e. strike price normalized by spot price or markets futures price) within which we calibrated our model.

$T_1^{\text{VIX}} = 0.0383$	$T_2^{\text{VIX}} = 0.0767$	$T_1^{\text{SPX}} = 0.0383$	$T_2^{\text{SPX}} = 0.1205$	$T_3^{\text{SPX}} = 0.1588$
(90%,220%)	(90%,220%)	(92%,105%)	(70%,105%)	(80%,120%)

The maturities in days are $\mathcal{T}^{\text{SPX}} = (14, 44, 58)$ and $\mathcal{T}^{\text{VIX}} = (14, 28)$. Let us also point out the high moneyness region (up to 220%) for VIX options is known to be challenging to fit. In our model, we need to choose both a primary process and a level of truncation of the signature a priori. Indeed, we treat these choices as *hyperparameters* and do not train them. As primary process we choose a three-dimensional OU process (see Example 2.9) with randomly chosen parameters

$$\kappa = (0.1, 25, 10)^\top, \quad \theta = (0.1, 4, 0.08)^\top, \quad \sigma = (0.7, 10, 5)^\top,$$

$$\rho = \begin{pmatrix} 1 & 0.213 & -0.576 & 0.329 \\ \cdot & 1 & -0.044 & -0.549 \\ \cdot & \cdot & 1 & -0.539 \\ \cdot & \cdot & \cdot & 1 \end{pmatrix}, \quad X_0 = (1, 0.08, 2)^\top.$$

and truncate the signature at $n = 3$. In this setting, our model has 85 trainable parameters, meaning that $\ell \in \mathbb{R}^{85}$. Note that choosing instead a two-dimensional Brownian motion as primary process does not lead to good fits at low truncation levels of the signature ($n = 3$), which is shown in [13, Appendix A]. The OU process is therefore a natural choice for a polynomial process which is tractable but exhibits richer dynamics than correlated Brownian motions alone. Moreover, OU processes have qualified for volatility modeling as shown in various articles, see e.g. [1, 2, 38, 40]. In terms of loss function, we choose the parameters $\lambda = 0.35$ and $\beta = 1$. For the evaluation of the pricing functional we simulate $N_{MC} = 80,000$ Monte Carlo samples.

The calibrated implied volatility smiles are depicted in Fig. 5 and the absolute value of the relative error between the models and markets future prices for each VIX-maturity are reported in the table below.

$T_1^{\text{VIX}} = 0.0383$	$T_2^{\text{VIX}} = 0.0767$
$\varepsilon_{T_1^{\text{VIX}}} = 9.8 \cdot 10^{-6}$	$\varepsilon_{T_2^{\text{VIX}}} = 6.6 \cdot 10^{-8}$

Indeed, the model does seem to fit the market implied volatilities of the VIX and SPX options at the given maturities rather well. As shown in Fig. 5, the model's implied volatilities lie within the market's bid-ask spreads and the calibrated future's prices coincide with the ones of the market. The latter is confirmed by the small relative errors reported in the table. However, it is also worth noting that this numerical experiment only consists of a small set of maturities and a single trading day. We refer the reader to [13] for a more thorough empirical study with calibration results for various sets of maturities and tests for parameter stability under re-calibration over multiple weeks.

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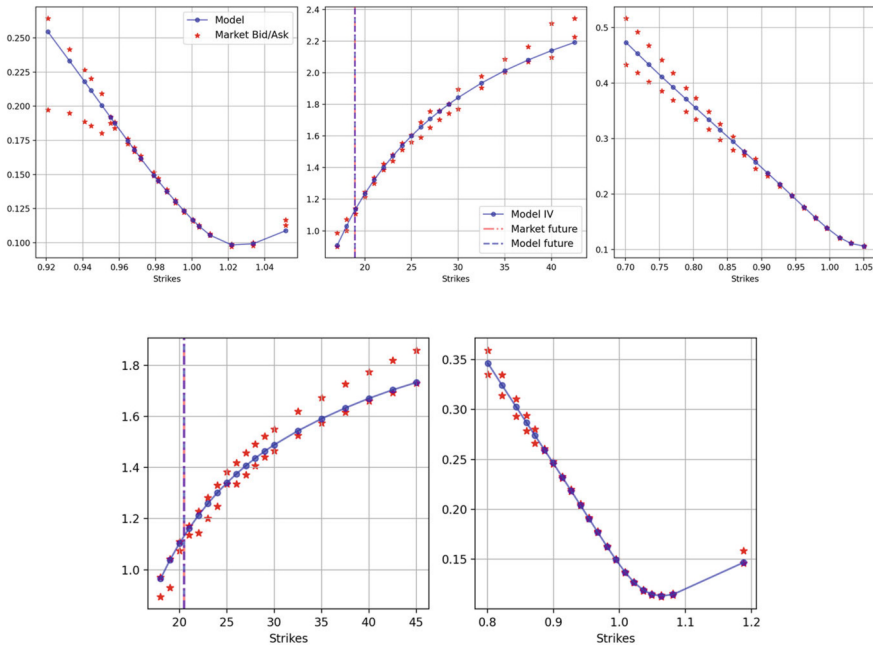


Fig. 5 The implied volatility smiles correspond to T_1^{SPX} , T_1^{VIX} , T_2^{SPX} , T_2^{VIX} , T_3^{SPX} (from top left to bottom right). The blue dots correspond to the values given by our calibrated model and the red stars are the markets bid-ask implied volatilities (from market call options). For the VIX, we additionally show the markets future price by the red dashed line and the one of the calibrated model by a blue dashed line

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